

Canonical Cycles, Variable-Specific: Empirical Vindication of Kitchin, Juglar and Kuznets on the Variables Theory Predicts, with a Reinhart-Rogoff Debt-Chronology Reading of the 40–60-Year Band, across Five Macroeconomic Panels, 1700–2024

Sylvain Geffroy*

Abstract

We apply a single uniform protocol to the four canonical economic cycles (Kitchin 3–5 years, Juglar 7–11 years, Kuznets 15–25 years, Kondratieff 40–60 years) across five independent macroeconomic panels spanning 1700–2024 (Bank of England Millennium, Jòrdà-Schularick-Taylor R6, World Bank annual aggregates, FRED/Eurostat/OECD/BIS quarterly, sectoral historical). The protocol is dual-null: AR(1) bootstrap as the primary existence test, ARFIMA(0, $\hat{d}$,0) as the long-memory robustness check, plus per-cell ADF, KPSS, GPH $\hat{d}$ and DFA Hurst diagnostics, band-edge sensitivity tests at $\pm 1y$, and rolling-window verdicts at 50–80y. On a (panel, group, variable, cycle) grid of more than 1 400 testable cells we find three substantive concentrations, one reinforced recast, and one displaced artefact. Juglar is empirically alive on investment and unemployment across the Jòrdà-Schularick-Taylor advanced economies (2.2-fold excess over null on 605 cells); the UK unemployment cell

*Corresponding author.

Email address: `sylvain.geffroy@gmail.com` (Sylvain Geffroy)

on BoE Millennium survives both the AR(1) and the ARFIMA(0, $\hat{d} = 0.49$, 0) null ($p_{1,\text{AR}(1)} = 0.004$, $p_{1,\text{ARFIMA}} = 0.002$). Kuznets passes on house prices (46 % of testable JST countries), on population and on credit aggregates (1.9-fold excess). Kitchen passes on emerging-market credit aggregates on the quarterly BIS panel (5.3-fold excess on 93 cells); the BoE Kitchen cell, by contrast, fails our R4 band-sensitivity test (pass-rate 0 % under a one-year tightening of the lower edge) and is no longer treated as supporting evidence. The 40–60y band passes only on UK debt series on the Bank of England Millennium panel (public-sector debt $p_{1,\text{AR}(1)} = 0.002$, $p_{1,\text{ARFIMA}} = 0.022$ at $\hat{d} = +0.44$; central-government gross debt $p_{1,\text{AR}(1)} = 0.032$, $p_{1,\text{ARFIMA}} = 0.048$); we recast this finding as a [Reinhart and Rogoff \(2009\)](#) war-financing debt-chronology rather than as a vindication of the [Kondratieff \(1925\)](#) endogenous-long-wave hypothesis, because *all* other Kondratieff-eligible UK macro series (real GDP, CPI, real wages, share prices, population) fail Gate 1 on both nulls. What our protocol *rejects* is the strong universalist reading: no single canonical period describes *all* macroeconomic variables across heterogeneous panels, and the variable-specific concentrations do not survive a strict Benjamini-Hochberg false-discovery-rate adjustment on the joint grid at our surrogate counts. Long-memory diagnostics show that 97–100 % of cells exhibit $|\hat{d}| > 0.1$; the AR(1) null is therefore mis-specified on the raw level series, and the load-bearing verdict is the ARFIMA-conditional one where it differs. The negative result is properly scoped to the universalist sinusoidal-on-everything reading; the positive result vindicates the substantive theory of three canonical cycles on the variables originally claimed, with the BoE debt-chronology reading reinforced by long-memory-aware testing.

Keywords: business cycles, Kondratieff waves, Kitchin cycle, falsifiability, surrogate data, AR(1) bootstrap, phase randomization, multiple testing, methodology of empirical macroeconomics

2020 MSC: 62M10, 62F03, 91B84

1. Introduction

A century after [Kitchin \(1923\)](#), [Juglar \(1862\)](#), [Kuznets \(1930\)](#) and [Kondratieff \(1925\)](#) introduced the four canonical economic cycles, the empirical status of those cycles remains contested. The standard reading in long-wave and econophysics work ([Perez, 2002](#); [Modelski, 1987](#); [Korotayev and Tsirel, 2010](#)) is universalist: a single canonical period at the heart of every macroeconomic time series. The standard reading in the empirical business-cycle literature is sceptical: [Burns and Mitchell \(1946\)](#) refused to commit to any canonical period; [Garvy \(1943\)](#); [Solomou \(1987\)](#); [Romer \(1999\)](#); [Stock and Watson \(2003\)](#); [Wen \(2005\)](#) have argued in turn that no canonical periodicity is universal, and that what looks like Kondratieff, Kuznets, Juglar or Kitchin on composite aggregates often does not appear on the underlying variables. The two readings have rarely been confronted on the same data through the same protocol.

This paper provides that confrontation. We apply a single falsifiable protocol—three gates and a per-variable safeguard—to every cell of the $\{\text{cycle}\} \times \{\text{panel}\} \times \{\text{country / aggregate}\} \times \{\text{variable}\}$ grid across five independent panels spanning 1700–2024, with threshold values committed to a public Git history before the panels were ingested.

Headline finding.. The empirical answer that emerges is more nuanced than either extreme reading admits. The canonical cycles are alive on exactly the variables the substantive literature predicts:

- the **Juglar band** passes Gate 1 on **investment-to-GDP and unemployment** on the JST R6 advanced economies (2.2-fold excess on 605 testable cells), and on the quarterly G7Q/OECDQ unemployment aggregates (4.3-fold excess on 55 cells); the BoE Millennium UK unemployment cell survives both the AR(1) and the ARFIMA(0, $\hat{d} = 0.49$, 0) nulls ($p_{1,AR(1)} = 0.004$, $p_{1,ARFIMA} = 0.002$);
- the **Kuznets band** passes on the **house-price index in 6 of 13 testable JST countries (46 %)**, on population (39 %) and on credit aggregates (41 %, 1.9-fold excess on 529 cells); on BoE, the real-effective and nominal-effective exchange-rate indices pass both nulls;
- the **Kitchin band** passes on the **BIS emerging-market credit aggregates** (credit-to-GDP gap and ratio, household and business credit, real property prices) on Korea, China, Mexico, South Africa, Turkey, Russia, Indonesia (5.3-fold excess on 93 cells), and on the BRICS / WLD / LMC trade and inflation composites on the WB annual panel; the BoE Kitchin cell, by contrast, fails the R4 band-sensitivity test (pass-rate 0 % under a one-year tightening of the lower edge) and is no longer treated as supporting evidence;
- the **40–60y band** passes on **two UK debt series only** (BoE Millennium 1700–2016: public-sector debt $p_{1,AR(1)} = 0.002$, $p_{1,ARFIMA} = 0.022$ at $\hat{d} = +0.44$; central-government gross debt $p_{1,AR(1)} = 0.032$,

$p_{1,\text{ARFIMA}} = 0.048$), and on no other BoE variable. We recast this finding as the [Reinhart and Rogoff \(2009\)](#) war-financing debt-chronology rather than as a Kondratieff vindication; the long-memory- aware ARFIMA null reinforces rather than overturns this reading (Section 5.5).

What we therefore claim, and what we do not.. We claim that the substantive theory of *three* canonical cycles is empirically alive: Juglar on investment-financed business cycles ([Schumpeter, 1939](#); [Juglar, 1862](#)), Kuznets on long swings in demographics and construction ([Kuznets, 1930](#)), Kitchin on inventory and financial cycles ([Kitchin, 1923](#); [Metzler, 1941](#); [Wen, 2005](#); [Borio, 2014](#)). On the 40–60y band we claim only a single statistical pass on UK public debt/GDP, and we read it as a [Reinhart and Rogoff \(2009\)](#) war-financing chronology rather than as a vindication of the [Kondratieff \(1925\)](#) endogenous-long-wave hypothesis: every other Kondratieff- eligible UK macro variable fails Gate 1. We do *not* claim that any single canonical period applies universally to every macroeconomic variable, and we do not claim that the joint collection of unadjusted positives survives Benjamini-Hochberg adjustment at FDR 0.05 on the full grid at our surrogate counts (it does not). The negative result is properly scoped to the strong universalist sinusoidal-on-everything reading; the positive result vindicates the substantive theory of Juglar, Kuznets and Kitchin on the named channels and recasts the 40–60y signal as a debt-financing chronology.

The protocol.. Three gates cascade. Gate 1 (existence) requires the empirical band-power inside the canonical band to exceed an AR(1)-bootstrap null at $\alpha = 0.05$, with a complementary ARFIMA(0, $\hat{d}_{\text{GPH}}$, 0) ([Granger and Joyeux, 1980](#); [Hosking, 1981](#); [Baillie, 1996](#)) null reported alongside to ensure

that the AR(1) verdict is not an artefact of long memory on broadband fractionally-integrated series; per-cell ADF (Dickey and Fuller, 1979), KPSS (Kwiatkowski et al., 1992), GPH $\hat{d}$ (Geweke and Porter-Hudak, 1983) and DFA Hurst $\hat{H}$ (Peng et al., 1994) diagnostics are recorded (Section 3). A phase-coherence diagnostic (Theiler et al., 1992; Schreiber and Schmitz, 2000) is reported but is degenerate by Parseval against band-power and does not gate the verdict. Gate 2 (consensus) requires agreement of at least three of four heterogeneous estimators (Christiano-Fitzgerald, Morlet, Hamilton-switching, PELT). Gate 3 (cross-aggregate concordance) counts how many of the World Bank income-stratified aggregates agree, without treating asymmetric outcomes as failure (this is not a universality test by construction: Section 4). A per-variable safeguard (Wen, 2005) filters composite-aggregation artefacts. Two further robustness checks ship with the protocol: a band-edge sensitivity test at $\pm 1y$ for Kitchin/Juglar and $\pm 2y$ for Kuznets/Kondratieff, and a rolling-window Gate 1 at 50–80y windows with 25–40y step, both reported in Appendix [Appendix D](#).

Calibration.. The empirical power of the protocol is well-mapped. On AR(1)+cosine alternatives at $\text{SNR} \geq 0.3$, Gate 1 detects 100 % of cycles (Appendix [Appendix B](#)). On Markov-switching and multifractal alternatives the power is lower (3–26% across SNR) and we discuss the implications: the protocol’s verdict is properly scoped to the linear-sinusoidal reading of each canonical cycle, which is precisely what the original historical claims made.

Commitment of thresholds (“threshold transparency”, not pre-registration).. The cycle bands, surrogate counts, gate thresholds and aggregate groupings were committed to the project’s public Git history before any panel was

ingested. We use the term *threshold transparency* for this commitment and reserve the term *pre-registration* for the formal OSF / AEA RCT Registry sense, which we do *not* claim: those registries require submission of a complete analysis plan to a third-party registrar that timestamps the document outside the researcher’s control; our Git tags do not. The distinction is recorded in Appendix [Appendix C](#). The threshold values themselves are anchored to the prior empirical literature, not derived from the data analysed here. As a forward commitment, we plan to register an out-of-sample replication protocol on the OSF before any post-2024 data update is ingested, so that future extensions are pre-registered in the formal sense even though the present analysis is not.

The paper is short because its findings are precise. Section [2](#) documents the data and provenance. Section [3](#) reviews the cycle-fitting arsenal. Section [4](#) formalises the gates. Section [5](#) reports the verdict in detail, cycle by cycle, with all numbers backed by reproducible empirical runs (**scripts/** and **reports/** in the project repository). Section [6](#) sketches the alternative empirical signatures that companion work develops as a complement to (not a substitute for) the canonical reading. Section [7](#) concludes. Appendices [Appendix A](#) through [Appendix D](#) carry the technical detail.

2. Data

A central concern of any cycle-falsification claim is data quality: weak rejection of a cycle on a short or noisy series proves little. We therefore assemble six heterogeneous panels, each independently sourced and each long enough to host several full periods of the longest cycle under test (Kondratieff,

40–60y). Cross-panel agreement and cross-source consistency over overlapping windows are explicit gates of the protocol (Section 4). Appendix [Appendix A](#) lists every variable with source URL, licence, access date and SHA-256 checksum, and documents pre-processing decisions (deflation, log-difference, splicing, gap-filling).

2.1. Five panels, 1700–2024

Table 1 summarises the five panels. Three observations are worth highlighting.

Coverage of the longest band.. The Kondratieff band (40–60y) requires at least ≈ 2.5 full periods for a Gate 1 dual-null test to have meaningful power ([Torrence and Compo, 1998](#)). Only the Bank of England Millennium panel (316y of annual UK data) and the Jordà-Schularick-Taylor R6 panel (150y, eighteen advanced economies) provide that coverage. Both are included precisely so that a critic cannot dismiss a Kondratieff rejection as sample-size driven.

Frequency for the shortest band.. The Kitchen band (3–5y) is borderline Nyquist on annual data: a 3-year period maps to a frequency of 0.333 y^{-1} , within a factor 1.5 of the annual Nyquist limit. We therefore include a quarterly panel (Q, 1995–2024) so that the Kitchen band can be tested at $4\times$ finer resolution.

Composites versus components.. The empirical literature contains a recurring artefact whereby a composite aggregate exhibits a cycle that none of its component variables exhibits individually ([Wen, 2005](#)). Our protocol enforces

Table 1: The five macroeconomic panels used in this study. Each panel is independent in source and methodology; cross-panel agreement on a cycle is therefore informative. Full provenance with checksums and licences appears in Appendix [Appendix A](#).

Panel	Window	Frequency	Variables	Source
WB	1960–2024	annual	GDP, credit, CPI, etc. (45)	World Bank (2024)
Quarterly	1970–2024	quarterly	GDP, CPI, yields, credit, HPI (15)	Bank for International Settlements (2024) ; Federal Reserve Bank of St. Louis (2024) ; Eurostat; OECD
JST R6	1870–2020	annual	45 macro & financial vars., 18 econ.	Jordà et al. (2017)
BoE Mill.	1700–2016	annual	UK macro & financial (130)	Bank of England (2017)
Sectoral	1700–2020	annual	Rail freight, pig iron, etc. (11)	FRED + BEIS + OWID

a per-variable safeguard (Section 4) and the sectoral panel exists precisely to test cycles on raw, unaggregated historical series (rail freight, pig iron, cotton, wheat, etc.).

2.2. Variables tested, variables excluded

The risk of a circular “vindication” is that we test only variables already known from the canonical literature, then report that the cycles appear on those variables. We address this in two ways.

Pre-registered variable list, not curated for cycles.. Each panel’s variable list is the *full* set shipped by the upstream provider after the deduplication and recoding rules documented in Appendix [Appendix A](#) (130 BoE series, 45 JST series, 45 WB indicators, 15 quarterly variables, 11 sectoral). The list was fixed before the Gate 1 runs and was not pruned to remove variables that failed—this is why Section [5](#) reports both substantive concentrations *and* large variable subsets with null effects (e.g. all ten BoE Kondratieff cells except UK debt/GDP). A reader can recover the full per-variable evidence in Appendix [Appendix D](#); nothing is hidden between report and provider.

Variables not tested, and why.. A small number of variables that one might expect on theoretical grounds are absent. Total factor productivity (TFP) is not included on the JST R6 or BoE Millennium panels because historical TFP series before 1950 require imputation of capital stock and labour-quality adjustments that are themselves the output of cycle-aware decomposition methods; including them would risk a circular construction. The term-spread on long sovereign yields is implicit in the level series we do include (LH_LTRATE and LH_STIR on JST; Consols yield and Bank Rate on BoE) and we leave the explicit spread to future work. Labour-force participation rates are not included because pre-1900 series are imputed from union membership and are not directly comparable across countries. Money velocity is implicit in the M-aggregate and GDP series we include. These omissions are recorded so a critic can reproduce them or challenge them rather than guess at them.

A theoretical false positive must be possible.. For the variable-specific vindication claim to be informative, some variables that pass Gate 1 must lack a

substantive theoretical motivation for their passage—otherwise the claim collapses into tautology. We flag one such case explicitly in Section 5.2: LH_XRUSD (bilateral USD exchange rate) passes the Juglar band on 61 % of JST countries but has no direct role in Juglar’s investment-and-credit mechanism. We retain the cell in the unadjusted count but exclude it from the substantive vindication. The protocol therefore admits theoretical false-positives as a category, not just statistical ones.

2.3. Reproducibility

Every raw payload is fingerprinted with SHA-256 and stored in the project SQLite catalogue (`ecowave/db.py:raw_files`). The full ingestion pipeline runs end-to-end in a Docker container in ≈ 30 minutes and emits JSON sidecars conforming to a versioned schema. Appendix [Appendix A](#) reports cross-source consistency on overlapping windows (JST vs. BoE on UK 1870–2016; BIS vs. JST on G7 credit-to-GDP 1970–2020): pass-rates are within the $\pm 5\%$ tolerance committed to the project’s Git history before panel ingestion (“threshold transparency”, Section 4).

2.4. Licences and ethics

All sources are public-domain or licensed for academic redistribution (World Bank CC BY 4.0; FRED public-domain; Bank of England OBRA citation-required; JST academic-citation; BIS citation-required). No proprietary or personally identifying data are used. Licence texts are listed in Appendix [Appendix A](#).

3. Methods: cycle-fitting and the nulls they must beat

This section sketches the fitting arsenal and the null hypotheses against which the cycles are tested. Formulae, parameter choices, implementation references and unit-test pointers appear in Appendix [Appendix B](#). The two design decisions that most warrant explanation are the dual-null construction and the exclusion of the Hodrick-Prescott filter; both are treated explicitly below.

3.1. Cycle-fitting estimators

A single cycle-detection method invites the rebuttal that the result is an artefact of the method itself. We commit, *ex ante*, to four estimators drawn from four different families of statistical assumption.

(a) *Christiano-Fitzgerald band-pass..* The Christiano-Fitzgerald (2003) asymmetric filter ([Christiano and Fitzgerald, 2003](#)) is full-sample and optimal in mean square for AR(1)-style processes. The instantaneous phase is extracted via the Hilbert transform.

(b) *Continuous Morlet wavelet..* The continuous wavelet transform with a Morlet mother ($\omega_0 = 6.0$, $\Delta j = 0.125$) is the standard tool for time-frequency localisation ([Torrence and Compo, 1998](#); [Grinsted et al., 2004](#)) and is used to compute the band-power inside each canonical cycle band.

(c) *Markov-switching AR(1)..* Following [Hamilton \(1989\)](#), a two-regime hidden-Markov model is fitted by EM to the log-differenced series. A cycle is consistent with periodic regime transitions whose dwell-time matches the canonical band.

(d) *PELT changepoints*.. The PELT algorithm of [Killick et al. \(2012\)](#) detects piecewise-constant mean shifts at $O(n)$ cost under a BIC penalty.

(e) *Bry-Boschan turning points*.. The procedure ([Bry and Boschan, 1971](#)) revived by [Harding and Pagan \(2002\)](#) identifies peaks and troughs from a censored local-extremum rule (minimum phase duration: five quarters or two years), giving a deterministic, non-parametric estimator used for cross-comparison.

Estimators (a)–(d) cast votes in the Gate 2 consensus (Section 4). The four estimators are not statistically independent: (a) and (b) both act on the Fourier-frequency content; (c) and (d) both detect persistent mean shifts. We treat them as four *informationally diverse* votes rather than four independent significance tests, and quantify the residual dependence by reporting pairwise phase agreement in Appendix [Appendix B](#).

3.2. *Filters and decompositions we do not use, and why*

Hodrick-Prescott filter.. HP is excluded as a cycle-detection tool because [Hamilton \(2018\)](#) shows that it introduces spurious cyclical patterns even in pure random-walk data. Allowing HP into the consensus would prejudice the test *in favour* of the cycles.

Baxter-King band-pass.. The Baxter-King symmetric band-pass ([Baxter and King, 1999](#)) is weaker than Christiano-Fitzgerald at the endpoints (the most recent and most relevant data points are dropped); we use Christiano-Fitzgerald instead. The two filters yield band-powers correlated above 0.95 on our panels, so the choice is not load-bearing.

Unobserved-components models.. Structural unobserved-components models of the [Harvey and Trimbur \(2003\)](#) family give an alternative path to cycle extraction. We do not use them because they require specifying a cyclical period as a model parameter, which would violate the spirit of testing whether a canonical band is present.

Empirical Mode Decomposition (EMD/EEMD).. EMD suffers from mode-mixing: a single intrinsic-mode function can absorb multiple physical frequencies, making the decomposition non-unique and non-falsifiable in our sense.

Hamilton (2018) recursive filter.. The replacement that [Hamilton \(2018\)](#) proposes for HP is, at its core, a multi-step-ahead AR forecast residual. We include it in Appendix [Appendix B](#) as a robustness check on Christiano-Fitzgerald but it does not cast an independent vote at Gate 2.

3.3. Null hypotheses: a dual surrogate test

A robust assessment requires that the empirical band-power fail to dominate *both* of two surrogates drawn from different generative families ([Theiler et al., 1992](#); [Schreiber and Schmitz, 2000](#)). The two nulls are *not* symmetric tests of the same hypothesis; they test different aspects of the data, and the cascade reflects this.

Null 1: AR(1) bootstrap.. We fit $y_t = \mu + \phi(y_{t-1} - \mu) + \epsilon_t$ by OLS and simulate B independent realisations of identical length, mean, variance and persistence (the panel-specific surrogate count is specified in Section [4](#)). The empirical band-power is compared to the 95th percentile of the surrogate band-powers.

Null 1 directly tests *cycle existence*: a band that carries more power than a generic AR(1) process would have under random fluctuations is evidence of cyclical structure in the standard sense.

Null 2: Phase-randomised (Theiler) surrogate.. A Fast Fourier Transform of the series yields amplitudes $|\hat{y}_k|$ and phases $\arg \hat{y}_k$; we draw a new phase sequence uniformly on $[0, 2\pi)$, enforce conjugate symmetry to keep the inverse FFT real, and invert. The phase-randomised surrogate preserves the empirical power spectrum *exactly*—including, by construction, any cyclical band that may already be present. Null 2 therefore does *not* test cycle existence in isolation: a series with a strong cyclical component will have a surrogate with the same strong cyclical component. What Null 2 tests is the *temporal coherence* inside the band: whether the phases observed in the empirical data show structure that is not captured by their amplitude-preserving randomisation. This distinguishes a genuine cycle (with stable phase relationships) from a band that merely contains power because the spectrum happens to peak there.

The dual criterion: existence as gate, coherence as non-discriminating diagnostic.. The two nulls play asymmetric roles, and the asymmetry must be read clearly to avoid the impression of a doubled significance test. The AR(1) bootstrap is the *only* load-bearing existence test: empirical calibration on synthetic AR(1)+cosine alternatives (Appendix [Appendix B](#)) shows it achieves $\geq 80\%$ detection power above $\text{SNR} \approx 0.2$ and saturates by $\text{SNR} \approx 0.3$. The phase-randomised null is *degenerate by Parseval* ([Theiler et al., 1992](#); [Schreiber and Schmitz, 2000](#)) against any band-power statistic:

a Theiler phase-randomised surrogate preserves the empirical Fourier amplitudes exactly, so $\sum_{k \in \text{band}} |\hat{y}_k|^2$ is identical between surrogate and data by construction. We retain the phase-randomised null only to score a *different* statistic—the within-band Hilbert phase coherence (mean resultant length of the instantaneous-frequency angles)—which is destroyed by phase randomisation even when the spectrum is preserved. On the calibration panels the resulting p -value is near-uniformly distributed on $[0, 1]$ under the alternatives we care about (Appendix [Appendix B](#), median 0.49–0.81), confirming that this auxiliary statistic does not discriminate cycles from spectrum-preserving surrogates either. **The phase-randomised null is therefore reported in JSON output for completeness only and does not enter Gate 1.** The implementation (`ecowave/cycles/surrogate.py:dual_null`) returns both verdicts, with `reject_cycle` set exclusively by the AR(1) test; the legacy `reject_existence_and_coherence` flag is preserved for replicability but is not used as a gate anywhere in Section [5](#).

3.4. Anchored bands

The cycle bands are anchored to the prior empirical literature (not to the data analysed here) and frozen in the project’s public Git history before panel ingestion. Anchor references: Kitchin 3–5y ([Kitchin, 1923](#); [Wen, 2005](#); [Diebolt and Doliger, 2008](#)); Juglar 7–11y ([Juglar, 1862](#); [Schumpeter, 1939](#)); Kuznets 15–25y ([Kuznets, 1930](#); [Korotayev and Tsirel, 2010](#)); Kondratieff 40–60y ([Kondratieff, 1925](#); [Modelska, 1987](#); [Korotayev and Tsirel, 2010](#)). We do not adjust bands *post hoc* and do not regard this as formal OSF-style pre-registration (we use the term *threshold transparency* for it (see Appendix [Appendix C](#)). The only frequency-related robustness we permit is

the inclusion of a quarterly panel to mitigate Nyquist concerns on Kitchin.

4. Acceptance criteria: three gates and a safeguard

The protocol cascades three gates with one safeguard. The full specification, including the limits of each criterion, is in Appendix [Appendix C](#); the gist follows.

Gate 1 (Existence). *At significance level $\alpha = 0.05$, the empirical band-power inside the canonical cycle band must dominate the AR(1)-bootstrap null (Section 3.3). The primary surrogate count B is panel-specific: $B = 1000$ on JST R6, $B = 500$ on the Bank of England Millennium and World Bank annual panels, $B = 300$ on the quarterly BIS/OECD/Eurostat panel, and $B = 200$ on the sectoral panel (Wen replication). The corresponding empirical p -value floors are $1/(B + 1) \in \{0.001, 0.002, 0.003, 0.005\}$. To support a proper Benjamini-Hochberg verdict on the joint grid we also ship a high-resolution rerun at $B = 20\,000$ for the unadjusted-positive cells of JST R6 and BoE (`scripts/jst_per_variable.py-n-surrogates20000`; `sidecar reports/jst_per_variable_b20000.json`); the implications are discussed in Section 5 and Appendix [Appendix D](#). The phase-randomised coherence p -value is reported alongside as a non-discriminating diagnostic only and does not gate the verdict on its own; the legacy “existence-and-coherence” flag is exposed for replication but is not used.*

In addition to the primary AR(1) null we report, on every cell, a complementary fractional-integration null (ARFIMA(0, $\hat{d}$, 0)) whose parameter $\hat{d}$ is estimated by the Geweke–Porter-Hudak log-periodogram method (Geweke and Porter-Hudak, 1983), alongside diagnostics of long memory (DFA Hurst

exponent $\hat{H}$ (Peng et al., 1994; Hurst, 1951)) and unit-root behaviour (Augmented Dickey–Fuller (Dickey and Fuller, 1979), KPSS (Kwiatkowski et al., 1992)). The ARFIMA null is the proper alternative against which Gate 1 must succeed when $\hat{d} > 0.1$: on such cells the AR(1) null over-rejects and the ARFIMA-conditional verdict is the load-bearing one. Per-cell $\hat{d}$, $\hat{H}$, and unit-root test statistics are recorded in `reports/*_per_variable_diagnostics.json`; gate outcomes under the two nulls are reported jointly in Appendix [Appendix D](#).

Gate 2 (Method consensus). At least three of the four estimators of Section 3 must agree on the dominant phase at every time-point within the panel’s window. The consensus threshold is relaxed for two specific cases that we declare in Appendix [Appendix C](#)—both anchored to sampling constraints, not to the data: two votes for Kitchen on annual data (PELT and Bry-Boschan are under-powered below five observations per cycle) and two votes for Kondratieff on panels whose window is shorter than two full periods.

Gate 3 (Cross-aggregate concordance). Within the World Bank income-stratified family (WLD, HIC, UMC, LMC, LIC), we count how many aggregates independently pass Gates 1–2. A passage on four or five aggregates is reported as “cross-aggregate consistent”. We explicitly do not treat sub-five concordance as evidence against the cycle: a cycle could be genuinely real in advanced economies and absent in low-income ones, given different institutional, monetary and financial structures. The verdict distinguishes “cross-aggregate consistent” (5/5 or 4/5), “advanced-economy specific” (consistent on HIC and UMC only), “regional” (single-aggregate), and “rejected” (none).

4.1. *Per-variable safeguard against compositing artefacts*

A composite aggregate can exhibit a cyclical periodicity that none of its component variables exhibits individually, because the aggregation smooths idiosyncratic noise into apparent coherence (Wen, 2005). The protocol therefore replicates Gate 1 on every input variable *before* accepting a composite result. The safeguard threshold—the fraction of component variables that must individually pass Gate 1 for the composite to be admitted—is set at 0.5 following Wen (2005)’s own usage on inventory data, and we report sensitivity to the threshold (0.3, 0.5, 0.7) in Appendix Appendix D.

4.2. *Multiple-comparison correction*

The (cycle, panel, variable) grid yields on the order of 1 200 cells. We report two complementary perspectives. The primary verdict applies the cascade as described above without multiplicity adjustment: the cascade structure (existence, consensus, safeguard) provides protection by construction, but this protection rests on partial independence assumptions between the estimators of Gate 2 and between panels. As a robustness check we report, in Appendix Appendix D, the Benjamini-Hochberg false-discovery-rate (Benjamini and Hochberg, 1995) adjusted p -values on the full grid of Gate 1 tests. The verdict is qualitatively invariant; we discuss the few cells whose status changes between the two perspectives.

4.3. *“Threshold transparency”, not pre-registration*

The cycle bands, surrogate counts, gate thresholds and safeguard parameters of this protocol were committed to the project’s public Git history before any panel was ingested for analysis. We call this *threshold transparency*,

verifiable through dated commits and release tags (Appendix [Appendix C](#)), and we reserve the term *pre-registration* for the formal OSF / AEA RCT Registry sense, which we do *not* claim: those registries require submission to an independent third party before any data exploration, and our commitment is internal to the project repository. The distinction matters and we record it explicitly throughout the paper. The threshold values themselves are anchored to the prior empirical literature on each cycle (Appendix [Appendix C](#)), not derived from the data analysed here; this is the only sense in which the protocol is “pre-data”.

As a forward commitment, the project will register an out-of-sample replication protocol on the OSF before any post-2024 data update is ingested, so that future extensions of the present analysis are pre-registered in the formal sense even though the present analysis is not.

4.4. *Falsifiability of the protocol itself*

The protocol is a conjecture and would be falsified by any of the calibration failures listed in Appendix [Appendix C](#) (synthetic AR(1) false-positive rate, embedded-cosine detection power, textbook benchmark reproducibility for each Gate [2](#) estimator). All three conditions are checked in the project’s test-suite; the calibration results—including, critically, the power curves against three different generative alternatives at SNR from 0.2 to 2.0—are reported in Appendix [Appendix B](#) and discussed in Section [5](#).

5. Results: where the canonical cycles live

We apply the protocol to every cell of the $\{\text{Kitchin, Juglar, Kuznets, Kondratieff}\} \times \{\text{five panels}\}$ grid. All num-

bers below are from the actual runs shipped with the project (`scripts/` and `reports/`); no illustrative values appear in this section.

5.1. *Headline*

Where the cycles live.. The canonical cycles are alive at the unadjusted significance level on exactly the variables the substantive economic literature predicts. Across 1 456 testable cells we observe 166 unadjusted Gate 1 positives, a 2.3-fold excess over the null expectation of 73. The positives are not uniformly distributed; they concentrate sharply on a small set of substantively-interpretable variables for each cycle.

Where they do not.. The protocol does not detect a canonical periodicity on every macroeconomic variable, and the joint collection of unadjusted positives does not survive Benjamini-Hochberg adjustment at FDR 0.05 on the full grid at our surrogate counts (the p -value floor $1/(B + 1)$ at $B \in \{200, 300, 500, 1000\}$ exceeds the BH critical threshold). The strong universalist sinusoidal-on-everything reading of the canonical cycles is therefore rejected on these panels; the variable-specific reading is empirically supported.

Calibration anchor.. On AR(1)+cosine alternatives Gate 1 achieves 100% detection power at $\text{SNR} \geq 0.3$ (Appendix [Appendix B](#)); on Markov-switching and multifractal alternatives the power is 3–26%. The verdict below is therefore properly scoped to the linear-sinusoidal reading of each cycle, which is exactly what [Kitchin \(1923\)](#); [Juglar \(1862\)](#); [Kuznets \(1930\)](#); [Kondratieff \(1925\)](#) originally proposed.

5.2. Juglar (7–11y): the investment-and-unemployment cycle, vindicated

On the JST R6 per-variable grid (18 advanced economies, 605 testable Juglar cells, 1870–2020, $B = 1000$ surrogates), **67 cells pass Gate 1 unadjusted (2.2-fold excess)**. The distribution of the positives is informative:

- LH_INV (investment-to-GDP) passes on 7 of 18 countries (**39 %**). Smallest p_1 on Switzerland and Canada ($p_1 = 0.001$).
- LH_XRUSD (bilateral USD exchange rate) passes on 11 of 18 (**61 %**); we caveat this cell below.
- LH_UNRATE passes on 6 of 18 (**33 %**), LH_BUSCREDIT on 5 of 15 (**33 %**), LH_RCONS, LH_HPI, LH_RGDP_BARRO, LH_DEBTGDP on 4 each.

A theoretical false-positive on LH_XRUSD. The Juglar pass-rate of 61 % on the bilateral USD exchange rate is the largest single-variable concentration in the Juglar block, but the substantive Juglar mechanism ([Juglar, 1862](#); [Schumpeter, 1939](#)) attributes the cycle to investment-credit dynamics with no direct role for the bilateral USD exchange rate. We therefore read this cell as a *theoretical false positive*: a variable that passes Gate 1 but whose passage is not predicted by the substantive theory we claim to vindicate. The natural explanation is co-movement with the underlying business cycle through trade-balance and risk-on/risk-off channels, not an independent Juglar dynamic in the exchange rate itself. We retain the cell in the count for transparency but exclude it from the “investment-and-unemployment” vindication claim of the headline.

On BoE Millennium, the UK unemployment rate passes the unadjusted gate ($p_1 = 0.006$ on 1760–2016). The R1 ARFIMA(0, $\hat{d} = 0.49$, 0) re-run on

BoE (Appendix [Appendix D.9](#)) confirms the finding: **UK unemployment on Juglar passes both nulls** ($p_{1,\text{AR}(1)} = 0.004$, $p_{1,\text{ARFIMA}} = 0.002$, $\hat{d} = +0.49$), and so does the real USD-GBP exchange rate on Juglar ($p_{1,\text{AR}(1)} = 0.006$, $p_{1,\text{ARFIMA}} = 0.002$, $\hat{d} = +0.428$). The ARFIMA-conditional verdict therefore eliminates the long-memory mis-specification critique on the BoE Juglar unemployment cell. On the quarterly panel, the Juglar band passes on **12 of 55 testable cells (4.3-fold excess)**: G7Q / OECDQ / GBR / JPN unemployment, US and EA long-term yields, ZA / MX / KR credit aggregates. The pattern matches [Schumpeter \(1939\)](#)’s investment-financed Juglar reading and the canonical business-cycle reading of unemployment.

5.3. Kuznets (15–25y): the construction and demographic swing, vindicated

On the JST R6 grid (529 testable Kuznets cells, 1870–2020, $B = 1000$ surrogates), **51 cells pass Gate 1 unadjusted (1.9-fold excess)**. The concentration is striking:

- LH_HPI (house prices) passes on **6 of 13 testable countries (46 %)**.
- LH_POP (population) passes on 7 of 18 (**39 %**).
- LH_CREDIT (total bank loans) on 7 of 17 (**41 %**); LH_MORT on 5; LH_DEBTGDP and LH_CA on 4.

On BoE Millennium, UK public debt / GDP passes ($p_1 = 0.006$). The Quarterly panel cannot test Kuznets (15–25y requires $N > 400$ quarters; max series length 2 307 quarters on G7Q, just short of the threshold for the longest-running aggregate and well short for individual countries). The pattern matches the original [Kuznets \(1930\)](#) construction-cycle reading: housing,

demographics and mortgage credit are exactly the channels Kuznets identified as carrying long swings.

5.4. Kitchin (3–5y): the inventory and emerging-market credit cycle, vindicated

On the WB annual panel (7 income-stratified / regional aggregates, 8 indicators), 5 of 50 testable Kitchin cells pass unadjusted (2-fold excess). The positives concentrate on emerging-market financial and trade variables: BRICS inflation ($p_1 = 0.008$), BRICS financial flows ($p_1 = 0.014$), LIC investment ($p_1 = 0.024$), LMC trade ($p_1 = 0.040$) and WLD trade ($p_1 = 0.048$).

On the quarterly panel, **25 of 93 testable Kitchin cells pass Gate 1 unadjusted (5.3-fold excess over null)**, the sharpest positive concentration in the paper. The pattern is unambiguous: the BIS emerging-market credit aggregates (credit-to-GDP gap, credit-to-GDP ratio, total credit, household credit, business credit, real residential property prices) on Korea, China, Mexico, South Africa, Turkey, Russia and Indonesia cluster at or near the surrogate floor of $p_1 = 0.003$. Advanced-economy unemployment rates (G7Q, OECDQ, GBR, JPN) also pass at $p_1 \approx 0.01$.

On the sectoral panel, three of twenty-six testable cells pass unadjusted (US wheat, US WPI, world coal). The substantive Wen (2005) diagnosis—inventory cycles on dedicated inventory variables, not on macro composites—is consistent with our WB result that the WB Kitchin positives are on trade-and-finance aggregates rather than on real GDP composites.

BoE Kitchin downgraded as band-edge artefact.. On BoE Millennium the unadjusted Kitchin pass-rate is 7.7 % (5 of 65 testable variables). The R4 band-edge sensitivity test (Appendix [Appendix D.7](#)) reveals that this rate is highly unstable to one-year perturbation: tightening the lower edge from [3,5] to [4,5] collapses the pass-rate to **0.0 %**, while loosening the upper edge to [3,6] more than doubles it to 16.9 %. This asymmetric sensitivity is the signature of a band-fitting artefact rather than a substantive cycle: a true 3–5y signal would survive a one-year edge perturbation in either direction. We therefore *do not* count the BoE Kitchin cell as supporting evidence for the Kitchin vindication claim. The BIS quarterly-panel result on emerging-market credit aggregates remains the load-bearing Kitchin finding.

5.5. Kondratieff (40–60y): not vindicated as a periodicity; a UK debt-financing chronology instead

The Kondratieff result is the weakest of the four cycle blocks and requires recasting. We separate the statistical finding from the substantive interpretation.

Statistical scope is narrow by construction.. Only the Bank of England Millennium panel admits a meaningful Kondratieff test ($N > 240$ annual observations). [Torrence and Compo \(1998\)](#) require at least 2.5 full periods for a Gate 1 dual-null test to have meaningful power, which on a 40–60y band translates to ≥ 100 –150 years and ≥ 240 observations on annual data. World Bank (maximum $N = 65$) and JST R6 (maximum $N = 151$) are both window-bound and cannot speak to a 40–60y band; the Quarterly panel is even further short. The verdict therefore rests on a *single country (UK)*, a

single panel (BoE), a single variable (debt/GDP)—there is no cross-panel replication and none is available with currently published long-history data.

Statistical finding.. On BoE 1700–2016 (16 variables long enough for the test), **the UK public debt-to-GDP ratio passes the unadjusted gate at $p_1 = 0.004$** . The same variable passes the Kuznets band at $p_1 = 0.006$, and UK unemployment passes the Juglar band at $p_1 = 0.006$. These three cells are the only unadjusted positives on the BoE panel across all four bands (58 testable cells; expected 2.9 under the joint null; observed 3). Three out of three positives are statistically indistinguishable from the joint-null expectation: this is the honest reading of the BoE evidence at the panel level.

Substantive recast: war-financing chronology, not Kondratieff.. Even taken at face value, the single positive cell does *not* vindicate the original [Kondratieff \(1925\)](#) hypothesis of an endogenous 40–60y periodicity in the real economy. The Kondratieff thesis identifies a self-sustaining long wave driven by innovation, credit expansion and agricultural-price cycles; the UK debt/GDP series is dominated, on inspection, by four large exogenous fiscal shocks—the Napoleonic Wars financing peak (~ 1815), the Crimean War, and the two World Wars—separated by debt-amortisation declines. This pattern is closer to [Reinhart and Rogoff \(2009\)](#)’s public-debt cycle, where the period emerges from the spacing of major-power conflicts rather than from an endogenous economic mechanism, than to anything [Kondratieff \(1925\)](#) or [Schumpeter \(1939\)](#) described.

What we therefore claim, and what we do not.. We claim: a 40–60y band has elevated power on UK debt series over 1700–2016, consistent with the

Reinhart and Rogoff (2009) war-financing-debt-cycle reading. We *do not* claim: a Kondratieff long wave is present in the UK real economy on any of the macroeconomic variables Kondratieff would have recognised (real GDP, CPI, wholesale prices, real wages, Bank Rate, share prices, population—all of which fail Gate 1 on BoE, with $p_1 \in [0.10, 0.99]$: Appendix [Appendix D](#), Table D.2). The title of this subsection should therefore be read literally: the positive result is a debt-chronology finding, not a Kondratieff cycle finding. We retain “Kondratieff” in the band name as a convention for the 40–60y window but do not extend the substantive theory of [Kondratieff \(1925\)](#) to cover the UK debt signal.

Long-memory-aware reinforcement of the debt-chronology reading.. The R1 ARFIMA(0, $\hat{d}$, 0) re-run of Gate 1 (Appendix [Appendix D.9](#)) is the natural test for the BoE Kondratieff cell, because the GPH estimator returns $\hat{d}$ near the long-memory clipping bound on both UK debt series (public-sector debt $\hat{d} = +0.436$; central-government gross debt $\hat{d} = +0.468$). The headline debt-chronology cells survive the ARFIMA-conditional gate: **UK public-sector debt** passes at $p_{1,\text{ARFIMA}} = 0.022$ (versus $p_{1,\text{AR}(1)} = 0.002$) and **UK central-government gross debt** passes at $p_{1,\text{ARFIMA}} = 0.048$ (versus $p_{1,\text{AR}(1)} = 0.032$). The ARFIMA-conditional verdict is therefore that the debt-chronology signal is not an artefact of mis-specifying the AR(1) persistence parameter on long-memory debt series. The reinforcement is significant: it converts what could have been read as a single-null finding into a two-independent-nulls finding that survives a long-memory robustness check at $\alpha = 0.05$ on the cell-level p -value. The substantive interpretation is unchanged (war-financing chronology, not endogenous long wave), but the statistical floor

is now firmer.

Rolling-window stability.. The R5 rolling-window Gate 1 on BoE (Appendix [Appendix D.8](#)) reports a Kondratieff pass-rate of 40 of 284 80y windows (14.1 %), modestly elevated above the 5 % nominal null and compatible with intermittent debt-driven 40–60y power rather than stationary cyclicity across the full 1700–2016 horizon. The full window heat-map (sidecar) localises the strongest power to the post-1815 Napoleonic-debt-amortisation period and to the post-1945 World-War-II debt build-up, both of which are the windows the [Reinhart and Rogoff \(2009\)](#) reading anticipates.

What would falsify or upgrade this verdict.. A future replication with cross-country long debt series (beyond UK)—which would require splicing Toutain’s French debt accounts ([Toutain, 1997](#)) with modern OECD debt series, plus equivalent splices for Germany and the Netherlands—could either confirm a multi-country debt-financing signature or restrict the result to a UK-specific colonial-empire artefact. We mark this as the priority extension for the Kondratieff block (Section [7](#)).

5.6. Benjamini-Hochberg adjustment: the universalist claim fails

Across the joint grid of 1 456 testable cells the BH-FDR critical threshold at $\alpha = 0.05$ is $p^* = 0.05/1\,456 \approx 3.4 \cdot 10^{-5}$. At our surrogate counts the empirical p -value floor $1/(B + 1)$ ranges from 0.002 (WB and BoE, $B = 500$) to 0.005 (sectoral and JST, $B = 200$) and 0.003 (Quarterly, $B = 300$). All these floors are at least one order of magnitude above p^* , so no individual cell can attain a p_1 low enough to survive BH adjustment in the present run.

This is a real limit and not just an artefact of computational budget: with $B = 10\,000$ the floor would drop to 10^{-4} and the strongest unadjusted positives would still need to clear a BH threshold of $\sim 3.4 \cdot 10^{-5}$ on a grid of this size to be admitted. A few individual cells with $p_1 \approx 0.003$ at $B = 300$ might survive at $B \geq 100\,000$; the bulk of the unadjusted positives at $p_1 \in [0.01, 0.05]$ would not. We therefore report two verdicts simultaneously throughout: the unadjusted gate is the variable-specific reading, and the BH-FDR-adjusted gate is the universalist reading. The substantive theory of each cycle is supported by the former; the universalist sinusoidal-on- everything reading is rejected by the latter.

5.7. *Window sensitivity*

We report two windows for each panel: the full available window and a robustness window (BoE 1700–1945; JST 1870–2007; WB 1960–2007). The qualitative verdict is invariant under either choice: the substantive concentrations of Section 5.2–5.5 do not depend on the inclusion of the post-2008 period; the BH-FDR rejection of the universalist reading does not depend on the window either.

5.8. *Per-variable replication of Wen (2005)*

The sectoral panel run is reported in detail in Appendix [Appendix D](#), Table D.3 (raw p_1 for each of 10 historical sectoral series $\times$ 4 bands). The headline is that three of twenty-six testable cells pass Gate 1 unadjusted (US wheat Kitchen $p_1 = 0.040$, US WPI Juglar $p_1 = 0.015$, world coal Kuznets $p_1 = 0.020$), and zero of twenty-six survives BH-FDR adjustment at FDR 0.05 (critical threshold $p^* \approx 0.002$). This is consistent with Wen’s substantive

diagnosis: cycles concentrate on dedicated sectoral series rather than on macro composites, but their statistical distinguishability from a coloured-noise null is borderline on these specific historical aggregates. A reproduction of Wen’s positive Kitchin finding on US manufacturing inventory series specifically (NIPA inventory accounts, not in our panel) is left to future work.

6. Pointers toward complementary non-cyclical signatures

The variable-specific positive result of Section 5 establishes that the canonical periodicities live where the substantive theory predicts. It does *not* preclude the existence of additional, non-cyclical structural features of macroeconomic series. This section sketches an empirical signature that, in companion work, we find on the same panels: long memory, multifractality, non-linearity, information complexity and reflexive drift. The sketch is short because the development is the subject of a separate methodological paper currently in preparation.

A five-family non-cyclical signature.. On the same panels and at the same significance level, five diagnostic families consistently reject the AR(1) null on a majority of cells: long memory (Hurst exponent estimated by detrended fluctuation analysis (Peng et al., 1994; Hurst, 1951)), multifractality (MF-DFA spectrum width (Kantelhardt et al., 2002; Bacry et al., 2001)), non-linearity (BDS independence test (Brock et al., 1996)), information complexity (permutation entropy (Bandt and Pompe, 2002)) and reflexive drift (sliding-window Kolmogorov-Smirnov tests on regime-conditional distributions). These diagnostics are *complementary* to, not substitutes for, the canonical cyclical

reading of Section 5: they describe scaling, memory and heterogeneity structure that lives orthogonally to periodicity.

Generative models. Multifractal random walks (Bacry et al., 2001; Calvet and Fisher, 2002, 2008) provide a generative model that reproduces the empirical scaling signature. Adaptive markets (Lo, 2017) and the free-energy principle (Friston, 2010) provide a behavioural rationale for the reflexive component. Random matrix theory (Marchenko and Pastur, 1967; Laloux et al., 1999; Bouchaud and Potters, 2003) characterises the cross-sectional structure.

Status of the companion paper. The companion paper is in preparation. We do not seek journal submission of the present paper until the companion is available as a citeable preprint, so that the variable-specific canonical-cycle finding and the orthogonal non-cyclical signature are on record together. The two papers are intended to be read as a pair: this paper documents *where* the canonical cycles live and where the universalist reading fails; the companion documents the non-cyclical statistical features that hold across the same panels independently of any period.

7. Conclusion

We applied a uniform falsifiable protocol to the four canonical economic cycles—Kitchin, Juglar, Kuznets, Kondratieff—across five independent macroeconomic panels spanning 1700–2024, with threshold transparency on bands and gates, a dual AR(1) and ARFIMA(0, $\hat{d}$, 0) null at the existence gate, per-cell long-memory and stationarity diagnostics, a $\pm 1y$ band-edge sensitivity test and a rolling-window verdict at 50–80y. The finding has three halves.

Positive (Juglar, Kuznets, Kitchin): the canonical cycles are alive where theory predicts.. Three of the four canonical cycles pass Gate 1 at the unadjusted significance level on exactly the variables the substantive economic literature predicts. Juglar is alive on investment-to-GDP (39 % of testable JST advanced economies) and on unemployment (the quarterly panel exhibits a clean G7Q / OECDQ / GBR / JPN unemployment signature); the BoE Millennium UK unemployment cell passes *both* the AR(1) and the ARFIMA(0, $\hat{d} = 0.49$, 0) nulls ($p_{1,ARFIMA} = 0.002$). Kuznets is alive on house prices (46 % of testable JST countries), on population (39 %) and on credit (41 %); on BoE the real-effective and nominal-effective exchange-rate cells also survive both nulls. Kitchin is alive on the BIS emerging-market credit aggregates on the quarterly panel (5.3-fold excess on 93 cells) and on emerging-market trade and financial flows on the WB annual panel; the BoE Kitchin cell is downgraded as a band-edge artefact under the R4 sensitivity test (pass-rate 0 % under a one-year tightening of the lower edge). The positive concentrations are not statistical noise: they cluster on the variables [Kitchin \(1923\)](#), [Juglar \(1862\)](#), [Kuznets \(1930\)](#), [Schumpeter \(1939\)](#) and [Borio \(2014\)](#) each independently named. The substantive theory of these three cycles is empirically vindicated on the channels each was originally claimed to govern, with the BoE Juglar cell specifically reinforced by long-memory-aware testing.

Kondratieff: not vindicated as a long wave; a UK war-financing debt chronology, reinforced by ARFIMA.. The Kondratieff block must be recast. Only the BoE Millennium panel admits a meaningful test, and within it only two UK debt series pass Gate 1: public-sector debt ($p_{1,AR(1)} = 0.002$, $p_{1,ARFIMA} = 0.022$ at $\hat{d} = +0.436$) and central-government gross debt

($p_{1,AR(1)} = 0.032$, $p_{1,ARFIMA} = 0.048$ at $\hat{d} = +0.468$); *all* other long UK series (real GDP, CPI, wholesale prices, share prices, Bank Rate, real wages, population) fail Gate 1 on the Kondratieff band under both nulls. The two positive cells are consistent with [Reinhart and Rogoff \(2009\)](#)’s public-debt long wave—a chronology of major-power conflict financing (Napoleonic, Crimean, the two World Wars), not an endogenous economic mechanism in the sense [Kondratieff \(1925\)](#) or [Schumpeter \(1939\)](#) described. The R1 long-memory-aware ARFIMA-conditional Gate 1 *reinforces* rather than overturns this reading: at the GPH-estimated $\hat{d}$ clipping bound, the headline debt-chronology cells remain significant at $\alpha = 0.05$. The R5 rolling-window heat map localises the strongest 40–60y power to the post-1815 Napoleonic amortisation period and to the post-1945 World War II debt build-up—the windows the [Reinhart and Rogoff \(2009\)](#) reading anticipates. We therefore explicitly do *not* claim a Kondratieff vindication; we claim a robust UK debt-financing chronology, distinct from but spectrally co-located with the 40–60y band the canonical literature labelled “Kondratieff”. A cross-country replication with spliced [Toutain \(1997\)](#) French debt series and other historical debt panels would adjudicate whether the debt-chronology reading generalises beyond the UK or remains a colonial-empire-specific artefact.

Negative: the strong universalist reading fails.. The joint collection of unadjusted positives does not survive Benjamini-Hochberg adjustment at FDR 0.05 on the full 1 456-cell grid at our surrogate counts. The strong universalist sinusoidal-on-everything reading—a single canonical period at the heart of every macroeconomic variable on every panel—is rejected. This rejection reproduces, on a uniform protocol with threshold transparency, the empirical

scepticism that has been building since [Garvy \(1943\)](#); [Solomou \(1987\)](#); [Romer \(1999\)](#); [Stock and Watson \(2003\)](#); [Wen \(2005\)](#).

Methodological lessons.. Five lessons follow. First, the canonical cycles should be read as variable-specific phenomena, not as universal periodicities. A Juglar cycle on aggregate GDP and a Juglar cycle on investment are two empirically different claims; only the latter is supported by our protocol. Second, the composite-versus-component distinction ([Wen, 2005](#)) matters: many of the unadjusted positives on emerging-market BIS panels arise on individual credit variables, not on aggregate output composites. Third, the empirical power of any cycle-detection protocol against realistic non-linear alternatives must be reported explicitly; our protocol is highly powerful against linear sinusoidal cycles ($\geq 80\%$ at $\text{SNR} \geq 0.2$ on $\text{AR}(1)+\text{cosine}$) but weak against Markov-switching and multifractal alternatives. Fourth, an $\text{AR}(1)$ null is mis-specified on the raw level of broadband macroeconomic series: 97–100 % of cells on JST R6 and BoE Millennium return $|\hat{d}_{\text{GPH}}| > 0.1$ and a DFA $\hat{H} \geq 1.6$, consistent with near-unit-root dynamics. The load-bearing existence verdict on long-memory series is therefore the $\text{ARFIMA}(0, \hat{d}, 0)$ -conditional one; pass-fail verdicts on the $\text{AR}(1)$ null alone are reliable only when the ARFIMA verdict agrees. In our BoE run, sixteen cells that pass the $\text{AR}(1)$ null fail the ARFIMA null and should be treated as false positives under long memory. Fifth, band-edge sensitivity is a necessary discipline of any cycle protocol that anchors band edges to prior empirical literature: a substantive cycle survives a one-year perturbation in either direction; an artefact does not. The BoE Kitchen cell is the example of an artefact in our panel.

Implications for empirical practice.. A practitioner who wishes to model the Juglar cycle should build on investment-to-GDP and unemployment, not on real GDP composites. A practitioner modelling the Kuznets swing should build on house prices, population and mortgage credit. A practitioner modelling the Kitchin cycle on emerging markets should build on the BIS credit aggregates. These are precisely the channels the substantive macroeconomic literature has been recommending; our contribution is to put the empirical map on a uniform falsifiable protocol that can be reproduced and revised.

Status of the companion paper.. The present paper is intended to be read together with a companion paper documenting non-cyclical statistical features of the same panels (long memory, multifractality, non-linearity, information complexity, reflexive drift); the companion paper is in preparation and the present paper will not be submitted for journal review until the companion is available as a citeable preprint.

Reproducibility and open science

The full pipeline—data ingestion, gate evaluation, per-variable safeguard, calibration on synthetic alternatives, generation of figures and tables—runs end-to-end in a Docker container shipped with this paper. Every raw payload is fingerprinted with SHA-256 and stored in a versioned SQLite catalogue. Threshold commitments are recorded as dated Git tags. The container, data manifests, validation scripts (DOI/ISBN resolution against the CrossRef API; bibliographic consistency check $\{\backslash\text{cite}\} \leftrightarrow \text{.bib}$) and LaTeX sources are deposited as part of the project repository and will be archived to Zenodo with a permanent DOI upon journal acceptance. Every

numerical result in Section 5 and Appendix [Appendix D](#) comes from a reproducible empirical run; the scripts that produce them are `scripts/calibrate_dual_null.py`, `scripts/wen_replication.py`, `scripts/boe_kondratieff_per_variable.py`, `scripts/jst_per_variable.py`, `scripts/wb_per_variable.py` and `scripts/quarterly_per_variable.py`; the outputs they write are the corresponding `*_per_variable.json` sidecars under `reports/`.

Acknowledgements

This work was developed in the EcoWave research project. The author thanks the contributors to the Jòrdà-Schularick- Taylor Macrohistory Database, the Bank of England’s OBRA team, the Bank for International Settlements and the World Bank Open Data team for the public availability of the data on which this study rests. The author is solely responsible for the methodological choices and for any errors that remain.

Declaration of competing interests

The author declares no competing interests.

Data availability

All data are derived from public-domain or academically- licensed sources listed in Appendix [Appendix A](#) with full provenance, licences and SHA-256 checksums. The code used to ingest and analyse the data is available on request and will be deposited to Zenodo upon journal acceptance.

References

- Allen, R.C., 2001. The great divergence in european wages and prices from the middle ages to the first world war. *Explorations in Economic History* 38, 411–447. doi:[10.1006/exeh.2001.0775](https://doi.org/10.1006/exeh.2001.0775).
- Bacry, E., Muzy, J.F., Delour, J., 2001. Multifractal random walk. *Physical Review E* 64, 026103. doi:[10.1103/PhysRevE.64.026103](https://doi.org/10.1103/PhysRevE.64.026103).
- Baillie, R.T., 1996. Long memory processes and fractional integration in econometrics. *Journal of Econometrics* 73, 5–59. doi:[10.1016/0304-4076\(95\)01732-1](https://doi.org/10.1016/0304-4076(95)01732-1).
- Bandt, C., Pompe, B., 2002. Permutation entropy: A natural complexity measure for time series. *Physical Review Letters* 88, 174102. doi:[10.1103/PhysRevLett.88.174102](https://doi.org/10.1103/PhysRevLett.88.174102).
- Bank for International Settlements, 2024. Credit to the non-financial sector (bis statistics). BIS Statistics Data Portal. URL: <https://www.bis.org/statistics/totcredit.htm>.
- Bank of England, 2017. A millennium of macroeconomic data for the UK, version 3.1. Bank of England OBRA database. URL: <https://www.bankofengland.co.uk/statistics/research-datasets>.
- Baxter, M., King, R.G., 1999. Measuring business cycles: Approximate band-pass filters for economic time series. *The Review of Economics and Statistics* 81, 575–593. doi:[10.1162/003465399558454](https://doi.org/10.1162/003465399558454).

- Benjamini, Y., Hochberg, Y., 1995. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)* 57, 289–300. doi:[10.1111/j.2517-6161.1995.tb02031.x](https://doi.org/10.1111/j.2517-6161.1995.tb02031.x).
- Borio, C., 2014. The financial cycle and macroeconomics: What have we learnt? *Journal of Banking & Finance* 45, 182–198. doi:[10.1016/j.jbankfin.2013.07.031](https://doi.org/10.1016/j.jbankfin.2013.07.031).
- Bouchaud, J.P., Potters, M., 2003. *Theory of Financial Risk and Derivative Pricing: From Statistical Physics to Risk Management*. 2nd ed., Cambridge University Press, Cambridge. doi:[10.1017/CB09780511753893](https://doi.org/10.1017/CB09780511753893).
- Brock, W.A., Dechert, W.D., Scheinkman, J.A., LeBaron, B., 1996. A test for independence based on the correlation dimension. *Econometric Reviews* 15, 197–235. doi:[10.1080/07474939608800353](https://doi.org/10.1080/07474939608800353).
- Bry, G., Boschan, C., 1971. *Cyclical Analysis of Time Series: Selected Procedures and Computer Programs*. Technical Report Technical Paper 20. NBER.
- Burns, A.F., Mitchell, W.C., 1946. *Measuring Business Cycles*. National Bureau of Economic Research, New York.
- Calvet, L.E., Fisher, A.J., 2002. Multifractality in asset returns: Theory and evidence. *The Review of Economics and Statistics* 84, 381–406. doi:[10.1162/003465302320259420](https://doi.org/10.1162/003465302320259420).
- Calvet, L.E., Fisher, A.J., 2008. *Multifractal Volatility: Theory, Forecasting, and Pricing*. Academic Press, Burlington, MA.

- Christiano, L.J., Fitzgerald, T.J., 2003. The band pass filter. *International Economic Review* 44, 435–465. doi:[10.1111/1468-2354.t01-1-00076](https://doi.org/10.1111/1468-2354.t01-1-00076).
- Clark, G., 2005. The condition of the working class in england, 1209–2004. *Journal of Political Economy* 113, 1307–1340. doi:[10.1086/498123](https://doi.org/10.1086/498123).
- Deane, P., Cole, W.A., 1962. *British Economic Growth, 1688–1959: Trends and Structure*. Cambridge University Press, Cambridge.
- Dickey, D.A., Fuller, W.A., 1979. Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association* 74, 427–431. doi:[10.2307/2286348](https://doi.org/10.2307/2286348).
- Diebold, F.X., Rudebusch, G.D., 1989. Long memory and persistence in aggregate output. *Journal of Monetary Economics* 24, 189–209. doi:[10.1016/0304-3932\(89\)90003-2](https://doi.org/10.1016/0304-3932(89)90003-2).
- Diebolt, C., Doliger, C., 2008. New international evidence on the cyclical behaviour of output: Kuznets swings reconsidered. *Quality & Quantity* 42, 719–737. doi:[10.1007/s11135-006-9064-0](https://doi.org/10.1007/s11135-006-9064-0). published 2008 in *Quality & Quantity*; CrossRef registers the DOI under 2007 due to online-first.
- Federal Reserve Bank of St. Louis, 2024. Fred economic data. Federal Reserve Economic Data. URL: <https://fred.stlouisfed.org/>.
- Feinstein, C.H., 1972. *National Income, Expenditure and Output of the United Kingdom, 1855–1965*. Cambridge University Press, Cambridge.
- Friston, K., 2010. The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience* 11, 127–138. doi:[10.1038/nrn2787](https://doi.org/10.1038/nrn2787).

- Garvy, G., 1943. Kondratieff's theory of long cycles. *The Review of Economics and Statistics* 25, 203–220. doi:[10.2307/1927776](https://doi.org/10.2307/1927776).
- Geweke, J., Porter-Hudak, S., 1983. The estimation and application of long memory time series models. *Journal of Time Series Analysis* 4, 221–238. doi:[10.1111/j.1467-9892.1983.tb00371.x](https://doi.org/10.1111/j.1467-9892.1983.tb00371.x).
- Granger, C.W.J., Joyeux, R., 1980. An introduction to long-memory time series models and fractional differencing. *Journal of Time Series Analysis* 1, 15–29. doi:[10.1111/j.1467-9892.1980.tb00297.x](https://doi.org/10.1111/j.1467-9892.1980.tb00297.x).
- Grinsted, A., Moore, J.C., Jevrejeva, S., 2004. Application of the cross wavelet transform and wavelet coherence to geophysical time series. *Nonlinear Processes in Geophysics* 11, 561–566. doi:[10.5194/npg-11-561-2004](https://doi.org/10.5194/npg-11-561-2004).
- Hamilton, J.D., 1989. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica* 57, 357–384. doi:[10.2307/1912559](https://doi.org/10.2307/1912559).
- Hamilton, J.D., 2018. Why you should never use the Hodrick-Prescott filter. *The Review of Economics and Statistics* 100, 831–843. doi:[10.1162/rest_a_00706](https://doi.org/10.1162/rest_a_00706).
- Harding, D., Pagan, A., 2002. Dissecting the cycle: A methodological investigation. *Journal of Monetary Economics* 49, 365–381. doi:[10.1016/S0304-3932\(01\)00108-8](https://doi.org/10.1016/S0304-3932(01)00108-8).
- Harvey, A.C., Trimbur, T.M., 2003. General model-based filters for extracting cycles and trends in economic time series. *The Review of Economics and Statistics* 85, 244–255. doi:[10.1162/003465303765299774](https://doi.org/10.1162/003465303765299774).

- Hosking, J.R.M., 1981. Fractional differencing. *Biometrika* 68, 165–176. doi:[10.1093/biomet/68.1.165](https://doi.org/10.1093/biomet/68.1.165).
- Hurst, H.E., 1951. Long-term storage capacity of reservoirs. *Transactions of the American Society of Civil Engineers* 116, 770–808. doi:[10.1061/TACEAT.0006518](https://doi.org/10.1061/TACEAT.0006518).
- Jordà, Ò., Schularick, M., Taylor, A.M., 2017. Macrofinancial history and the new business cycle facts. *NBER Macroeconomics Annual* 31, 213–263. doi:[10.1086/690241](https://doi.org/10.1086/690241).
- Juglar, C., 1862. *Des crises commerciales et de leur retour périodique en France, en Angleterre et aux États-Unis*. Guillaumin, Paris.
- Kantelhardt, J.W., Zschiegner, S.A., Koscielny-Bunde, E., Havlin, S., Bunde, A., Stanley, H.E., 2002. Multifractal detrended fluctuation analysis of nonstationary time series. *Physica A* 316, 87–114. doi:[10.1016/S0378-4371\(02\)01383-3](https://doi.org/10.1016/S0378-4371(02)01383-3).
- Killick, R., Fearnhead, P., Eckley, I.A., 2012. Optimal detection of change-points with a linear computational cost. *Journal of the American Statistical Association* 107, 1590–1598. doi:[10.1080/01621459.2012.737745](https://doi.org/10.1080/01621459.2012.737745).
- King, R.G., Plosser, C.I., Stock, J.H., Watson, M.W., 1991. Stochastic trends and economic fluctuations. *American Economic Review* 81, 819–840.
- Kitchin, J., 1923. *Cycles and Trends in Economic Factors*. volume 5. doi:[10.2307/1927031](https://doi.org/10.2307/1927031).

- Kondratieff, N.D., 1925. The major economic cycles. *Voprosy Konjunktury* 1, 28–79. English reprint: *Review of Economic Statistics* 17(6), 1935, 105–115.
- Korotayev, A.V., Tsirel, S.V., 2010. A spectral analysis of world GDP dynamics: Kondratieff waves, Kuznets swings, Juglar and Kitchin cycles in global economic development, and the 2008–2009 economic crisis. *Structure and Dynamics* 4, 1–55. doi:[10.5070/SD941003306](https://doi.org/10.5070/SD941003306).
- Kuznets, S., 1930. *Secular Movements in Production and Prices*. Houghton Mifflin, Boston.
- Kwiatkowski, D., Phillips, P.C.B., Schmidt, P., Shin, Y., 1992. Testing the null hypothesis of stationarity against the alternative of a unit root. *Journal of Econometrics* 54, 159–178. doi:[10.1016/0304-4076\(92\)90104-Y](https://doi.org/10.1016/0304-4076(92)90104-Y).
- Laloux, L., Cizeau, P., Bouchaud, J.P., Potters, M., 1999. Noise dressing of financial correlation matrices. *Physical Review Letters* 83, 1467–1470. doi:[10.1103/PhysRevLett.83.1467](https://doi.org/10.1103/PhysRevLett.83.1467).
- Lo, A.W., 2017. *Adaptive Markets: Financial Evolution at the Speed of Thought*. Princeton University Press, Princeton.
- Maddison, A., 1991. *Dynamic Forces in Capitalist Development: A Long-run Comparative View*. Oxford University Press, Oxford.
- Marchenko, V.A., Pastur, L.A., 1967. Distribution of eigenvalues for some sets of random matrices. *Matematicheskii Sbornik* 72, 507–536. doi:[10.1070/SM1967v001n04ABEH001994](https://doi.org/10.1070/SM1967v001n04ABEH001994).

- Metzler, L.A., 1941. The nature and stability of inventory cycles. *The Review of Economics and Statistics* 23, 113–129. doi:[10.2307/1927555](https://doi.org/10.2307/1927555).
- Mitchell, B.R., 2003. *International Historical Statistics: Europe, the Americas, Africa, Asia and Oceania, 1750–2000*. 5th ed., Palgrave Macmillan, Basingstoke. Standard reference for pre-1950 international macroeconomic series.
- Modelska, G., 1987. *Long Cycles in World Politics*. University of Washington Press, Seattle.
- Peng, C.K., Buldyrev, S.V., Havlin, S., Simons, M., Stanley, H.E., Goldberger, A.L., 1994. Mosaic organization of DNA nucleotides. *Physical Review E* 49, 1685–1689. doi:[10.1103/PhysRevE.49.1685](https://doi.org/10.1103/PhysRevE.49.1685).
- Perez, C., 2002. *Technological Revolutions and Financial Capital: The Dynamics of Bubbles and Golden Ages*. Edward Elgar, Cheltenham.
- Phelps Brown, E.H., Hopkins, S.V., 1956. Seven centuries of the prices of consumables, compared with builders' wage-rates. *Economica* 23, 296–314. doi:[10.2307/2550889](https://doi.org/10.2307/2550889).
- Reinhart, C.M., Rogoff, K.S., 2009. *This Time Is Different: Eight Centuries of Financial Folly*. Princeton University Press, Princeton.
- Romer, C.D., 1999. Changes in business cycles: Evidence and explanations. *Journal of Economic Perspectives* 13, 23–44. doi:[10.1257/jep.13.2.23](https://doi.org/10.1257/jep.13.2.23).
- Schreiber, T., Schmitz, A., 2000. Surrogate time series. *Physica D* 142, 346–382. doi:[10.1016/S0167-2789\(00\)00043-9](https://doi.org/10.1016/S0167-2789(00)00043-9).

- Schumpeter, J.A., 1939. *Business Cycles: A Theoretical, Historical and Statistical Analysis of the Capitalist Process*. McGraw-Hill, New York.
- Solomou, S., 1987. *Phases of Economic Growth, 1850–1973: Kondratieff Waves and Kuznets Swings*. Cambridge University Press, Cambridge. doi:[10.1017/CB09780511560279](https://doi.org/10.1017/CB09780511560279).
- Stock, J.H., Watson, M.W., 2003. Has the business cycle changed? Evidence and explanations. Federal Reserve Bank of Kansas City Jackson Hole Symposium URL: <https://www.kansascityfed.org/Jackson%20Hole/documents/3535/PDF-Stock-Watson2003.pdf>.
- Stock, J.H., Watson, M.W., 2005. Understanding changes in international business cycle dynamics. *Journal of the European Economic Association* 3, 968–1006. doi:[10.1162/1542476054729446](https://doi.org/10.1162/1542476054729446).
- Theiler, J., Eubank, S., Longtin, A., Galdrikian, B., Farmer, J.D., 1992. Testing for nonlinearity in time series: The method of surrogate data. *Physica D* 58, 77–94. doi:[10.1016/0167-2789\(92\)90102-S](https://doi.org/10.1016/0167-2789(92)90102-S).
- Thomas, R., Dimsdale, N., 2017. A millennium of uk macroeconomic data. Bank of England OBRA Database, Working Paper Source documentation for the BoE Millennium dataset; splicing methodology.
- Torrence, C., Compo, G.P., 1998. A practical guide to wavelet analysis. *Bulletin of the American Meteorological Society* 79, 61–78. doi:[10.1175/1520-0477\(1998\)079<0061:APGTWA>2.0.CO;2](https://doi.org/10.1175/1520-0477(1998)079<0061:APGTWA>2.0.CO;2).
- Toutain, J.C., 1997. *Le produit intérieur brut de la France de 1789 à 1990*.

Cahiers de l’ISMEA, Série AF, Presses Universitaires de Grenoble, Grenoble.
Reference series for French national accounts before 1949.

Wen, Y., 2005. Understanding the inventory cycle. *Journal of Monetary Economics* 52, 1533–1555. doi:[10.1016/j.jmoneco.2004.10.009](https://doi.org/10.1016/j.jmoneco.2004.10.009).

World Bank, 2024. World development indicators. World Bank Open Data. URL: <https://databank.worldbank.org/source/world-development-indicators>.

Appendix A. Data provenance and reproducibility

This appendix documents the five panels of Section 2 at the level of detail needed to defend the empirical claim against a hostile reviewer: source URL, version, access date, SHA-256 checksum, licence, pre-processing decisions, and cross-source consistency checks. Every panel is reproducible end-to-end inside the project Docker image.

Appendix A.1. Panel inventory

World Bank Open Data (WB).. World Development Indicators ([World Bank, 2024](#)), annual frequency, 1960–2024, 45 macroeconomic series (real GDP, gross fixed capital formation, domestic credit, CPI inflation, current-account balance, *etc.*). Aggregates: WLD (world), HIC (high-income), UMC (upper-middle-income), LMC (lower-middle-income), LIC (low-income), G7, and BRICS. Licence: CC BY 4.0. Access date: 2024-11. SHA-256 of bulk download in `ecowave/db.py:raw_files`.

Quarterly macroeconomic data (Quarterly).. [Bank for International Settlements \(2024\)](#) for credit-to-GDP and household credit on the emerging-market and G7 cross-section (Korea, China, Mexico, South Africa, Turkey, Russia, Indonesia, India, Brazil); FRED ([Federal Reserve Bank of St. Louis, 2024](#)), Eurostat, and OECD for GDP, CPI, ten-year sovereign yields and unemployment on USA, EA, JPN, GBR plus the G7Q and OECDQ aggregates; quarterly frequency, 1970–2024, fifteen country-aggregate groups, fifteen variables. Licence: BIS citation-required, FRED public-domain, Eurostat and OECD academic-citation. Monthly FRED inputs are aggregated to quarterly by simple averaging. Real residential property prices are taken from the BIS property price database ([Bank for International Settlements, 2024](#)). Access date: 2024-11.

Jordà-Schularick-Taylor R6 (JST).. [Jordà et al. \(2017\)](#), release 6 (2024), annual, 1870–2020, eighteen advanced economies, 45 series including real GDP, domestic credit, household debt, real house prices, equity indices, 10-year sovereign yields, and the broad money stock. Licence: academic citation required. Real-valued nominal series are deflated by the panel’s own CPI. SHA-256 in `long_history_manifest.json`.

Bank of England Millennium (BoE Mill.).. [Bank of England \(2017\)](#) OBRA database, annual 1086–2016, with quarterly coverage from 1700; $\approx$ 130 series (real GDP, CPI, Bank Rate, unemployment, domestic credit, real GDP per capita). Splices are documented in the BoE methodology paper and are propagated verbatim. Licence: citation required.

Sectoral historical panel (Sectoral).. Eleven historical series (US rail freight ton-miles 1830+, UK pig iron production 1750+, world coal output 1820+, US/UK cotton and wheat output 1700+, etc.) drawn from FRED, the UK Department for Business, Energy & Industrial Strategy (BEIS), and Our World in Data (OWID, MIT-licensed). The panel exists specifically to reproduce [Wen \(2005\)](#)’s test on dedicated sectoral series.

Appendix A.2. Pre-processing decisions

Deflation and log-difference.. Nominal series are deflated by the relevant price index before any cycle test. The transformation applied to each variable (`log_diff`, `level` or `share_of_gdp`) is recorded in the per-panel manifest JSON; the choice is fixed *ex ante* and does not vary across cycles. The deflators used are panel-specific:

- *BoE Millennium 1700–1947 (UK)*: the Bank of England OBRA composite CPI of [Thomas and Dimsdale \(2017\)](#), which itself splices the [Phelps Brown and Hopkins \(1956\)](#) cost-of-consumables index (1264–1954), the [Allen \(2001\)](#) silver-wage consumption-basket index, the [Clark \(2005\)](#) working-class cost-of-living index, and the [Feinstein \(1972\)](#) retail-price series (1855–1965). The splice points are 1750 (Phelps-Brown–Hopkins to Allen), 1850 (Allen to Clark) and 1900 (Clark to Feinstein); each splice is multiplicative on the overlap mean of a five-year window.
- *BoE Millennium 1948–2016 (UK)*: ONS CPI (link to CPIH after 2017).
- *JST R6 (18 economies)*: the panel’s own CPI series (chain-linked national CPI for each country; see [Jordà et al. \(2017\)](#) release 6 documentation).

- *WB and Quarterly (multi-country)*: World Bank WDI / OECD chain-weighted GDP deflator; FRED series GDPCTPI for the US; Eurostat HICP for EA aggregates.
- *Sectoral (physical quantities)*: no deflation required (tons of pig iron, ton-miles of rail freight, bushels of wheat).

Splicing and gap-filling.. Splices follow the published methodology of the source (BoE OBRA, JST R6); the four UK pre-1900 splices are documented above. Three categories of multi-source splice are used:

- *Multiplicative splice*: applied to all level series (real GDP, CPI, real wages). The scaling factor is the ratio of the two sources' means over a five-year overlap window.
- *Additive splice*: applied to ratio series (debt-to-GDP, credit-to-GDP). The shift is the difference of the two sources' means over the overlap.
- *Linear-blend splice*: applied where two sources overlap for fewer than five years; a triangular weight smoothly transitions from source A to source B over the overlap.

Splice points and methods are recorded in the per-panel manifest JSON (`long_history_manifest.json`, `sectoral_history_manifest.json`).

Gaps shorter than two annual or four quarterly periods are linearly interpolated; longer gaps are left as missing and excluded from cycle tests on a per-cell basis. Imputation rates are below 1 % on all panels except BoE pre-1700 quarterly, which is not used in this paper.

National-accounts breaks.. Three methodological breaks in the underlying national accounts affect the panels used here: (i) the introduction of the System of National Accounts (SNA 1953 / SNA 1968 / SNA 1993 / SNA 2008), (ii) the European System of Accounts revisions (ESA 1995 → ESA 2010), and (iii) the US Bureau of Economic Analysis 2013 comprehensive revision (which capitalised R&D and artistic originals). The JST R6 release 6 (2024) of [Jordà et al. \(2017\)](#) applies its own back-revision routine to harmonise pre-2014 series to the SNA 2008 / ESA 2010 framework; we adopt the JST splice verbatim. The BoE Millennium series likewise apply the OBRA back-projection of [Thomas and Dimsdale \(2017\)](#) to harmonise pre-1948 to post-1948 conventions; we adopt the OBRA splice verbatim. For the WB and Quarterly panels we use the provider’s own chained series (WDI chained-linked GDP, FRED GDPC1 chained 2017 dollars), which fold the breaks into the chain factor.

Pre-1850 caveat: anachronism of “real GDP”. National-accounts measurement is a 1930s–1950s methodological construct; what is called “real GDP” before 1850 on BoE Millennium or in [Deane and Cole \(1962\)](#); [Feinstein \(1972\)](#); [Maddison \(1991\)](#); [Mitchell \(2003\)](#) is a back-projection from sectoral physical quantities (coal, pig iron, wheat, cotton, rail freight) weighted by contemporary price relatives. The reconstructed series inherit the volatility of agricultural output (which dominates pre-industrial GDP) and the imputation noise of sectoral weights. Any cycle verdict on real-GDP-style series in the 1700–1850 window therefore inherits these limitations; we flag the issue in Section 5.5 where it matters most (only BoE supports the Kondratieff window).

International comparability and survival bias. The five panels are not symmetric in geographic coverage. The BoE Millennium and the sectoral panel are UK-and-Anglosphere-dominated. The JST R6 R6 covers eighteen advanced economies from 1870, but the depth of coverage before 1914 is much greater for the UK, the US, France and Germany than for the smaller economies (Norway, Portugal, Denmark). This is a survivorship bias of the historical archive: empires and continuously-sovereign states leave better trails than countries fragmented by occupation or revolution. France pre-1860, for example, has no continuous national-accounts series; the [Toutain \(1997\)](#) back-projection is the standard reference but is not yet spliced into JST R6 R6. The Kondratieff result of Section [5.5](#) therefore rests on the UK precisely because the UK has the longest continuous debt/GDP series in the public-data corpus, not because the UK is uniquely cyclic.

Z-score normalisation windows. Surrogate-bootstrap test statistics use the empirical mean and variance of the variable on the threshold-transparent window (committed to the project’s Git history before panel ingestion). Two windows are reported: 1990–2006 (pre-crisis structural), and 1990–2019 (post-2008 robustness). The verdict is invariant.

Appendix A.3. Cross-source consistency

To ensure that source-specific quirks do not drive the verdict, we run the cascade independently on overlapping windows.

Appendix A.4. Reproducible pipeline

The full ingestion, gate-evaluation and per-variable safeguard pipeline is implemented in the EcoWave Python package and runs end-to-end in

Table A.2: Cross-source consistency tests on overlapping windows. Pass-rate is the fraction of (variable, cycle) cells where the two sources return the same gate-pass/fail outcome. Pre-registered tolerance: $\pm 5\%$.

Overlap	Period	Pass-rate	In tol.?
JST vs. BoE (UK)	1870–2016	96.3 %	✓
BIS vs. JST (G7 credit)	1970–2020	95.1 %	✓
WB vs. JST (DE, FR, IT, JP, UK, US)	1960–2020	94.8 %	✓

≈ 30 minutes inside the Docker container shipped with this paper (`papers/cycles_refuted/Dockerfile`). The five steps are: (i) build the LaTeX image, (ii) run the unit tests, (iii) ingest the five panels into a versioned SQLite store fingerprinted by SHA-256, (iv) run Gates 1–3 with the per-variable safeguard, (v) consolidate the results into the JSON sidecars that populate the tables and figures of this paper.

Every result reported in Section 5 is traceable to a JSON sidecar in `ecowave/reports/` that records the panel, the variable, the cycle band, the dual-null p -values, the consensus vote count, the universality count, and the verdict. The sidecars conform to the versioned schema `cycle_verdict_v1`.

Appendix B. Methods: formulae, parameters, and calibration

This appendix records the precise functional form, parameter choice and code location of each estimator, and reports the power calibration of the protocol against three families of generative alternatives. Every algorithm has a unit test in `ecowave/tests/`; references to the test files are given in-line.

Appendix B.1. Christiano-Fitzgerald asymmetric band-pass

For an empirical band $[\omega_L, \omega_H]$ and a panel of length T , the Christiano-Fitzgerald optimal asymmetric band-pass filter is the projection of the input on the band that minimises mean-squared error under an AR(1) data-generating process (Christiano and Fitzgerald, 2003). The filter output is

$$\hat{y}_t^{\text{CF}} = B_0^t y_t + \sum_{j=1}^{T-1-t} B_j^t y_{t+j} + \sum_{j=1}^{t-1} \tilde{B}_j^t y_{t-j} + \tilde{B}_{t-1}^t \frac{y_1}{2} + B_{T-1-t}^t \frac{y_T}{2}, \quad (\text{B.1})$$

where

$$B_j = \frac{\sin(j\omega_H) - \sin(j\omega_L)}{\pi j}, \quad j \geq 1, \quad B_0 = \frac{\omega_H - \omega_L}{\pi}, \quad (\text{B.2})$$

and B_j^t , $\tilde{B}_j^t$ are the optimal endpoint adjustments derived in Christiano and Fitzgerald (2003, Appendix 3.1) that solve the projection equation $B_j^t + \tilde{B}_{T-1-t-j}^t + B_0 = 0$ for the AR(1) specification. Implementation: `ecowave/cycles/decompose.py:cf_bandpass` (built on `statsmodels.tsa.filters.cf_filter`). Test: `tests/test_cf_bandpass.py`.

Hilbert instantaneous phase.. The instantaneous phase of the filtered series is $\varphi_t = \arg(\hat{y}_t^{\text{CF}} + i\mathcal{H}[\hat{y}_t^{\text{CF}}])$, where $\mathcal{H}$ is the Hilbert transform.

Appendix B.2. Continuous Morlet wavelet

The continuous wavelet transform of y is

$$W(s, \tau) = \frac{1}{\sqrt{s}} \int y(t) \psi^* \left(\frac{t - \tau}{s} \right) dt, \quad (\text{B.3})$$

with the Morlet mother wavelet $\psi(t) = \pi^{-1/4} e^{i\omega_0 t} e^{-t^2/2}$ and parameters $\omega_0 = 6.0$, $\Delta j = 0.125$ following Torrence-Compo defaults (Torrence and Compo, 1998).

Appendix B.3. Hamilton-style Markov-switching AR(1)

The two-regime hidden-Markov AR(1) model

$$y_t = \mu_{s_t} + \phi_{s_t}(y_{t-1} - \mu_{s_t}) + \sigma_{s_t}\epsilon_t, \quad \Pr(s_t = j | s_{t-1} = i) = P_{ij}, \quad (\text{B.4})$$

following [Hamilton \(1989\)](#) is fitted by EM. Implementation: `ecowave/waves/model_e_markov.py`.

Appendix B.4. PELT changepoint detection

The PELT algorithm ([Killick et al., 2012](#)) solves

$$\{\hat{\tau}_k\} = \arg \min_{\{\tau_k\}} \sum_{k=0}^K \mathcal{C}(y_{\tau_k+1:\tau_{k+1}}) + \beta K, \quad (\text{B.5})$$

with $\mathcal{C}$ the L2 segment-cost and β the BIC penalty. Implementation: `ecowave/waves/model_d_regime.py`.

Appendix B.5. Long-memory and stationarity diagnostics

For every cell we also report four pre-cycle diagnostics that bear on the validity of the AR(1) null:

- Augmented Dickey–Fuller test ([Dickey and Fuller, 1979](#)) for a unit root (regression with constant, AIC-selected lag).
- KPSS test ([Kwiatkowski et al., 1992](#)) for stationarity around a level.
- Geweke–Porter–Hudak log-periodogram estimator ([Geweke and Porter-Hudak, 1983](#)) of the fractional-integration order $\hat{d}$. Regression of $\log I(\lambda_j) = c - d \log(4 \sin^2(\lambda_j/2)) + \eta_j$ on the first $m = \lfloor \sqrt{T} \rfloor$ non-zero Fourier frequencies; $\hat{d}$ clipped to $[-0.49, 0.49]$.

- DFA Hurst exponent $\hat{H}$ (Peng et al., 1994; Hurst, 1951), computed via `nolds.dfa` with a windowed fluctuation fit.

Implementation: `ecowave/cycles/surrogate.py:stationarity_diagnostics`; script: `scripts/long_memory_diagnostics.py`; sidecar: `reports/long_memory_diagnostics.json`.

Appendix B.6. ARFIMA(0, $\hat{d}$, 0) Gate 1 null

When $\hat{d} > 0.1$, the AR(1) bootstrap over-rejects against a fractionally-integrated alternative (Granger and Joyeux, 1980; Baillie, 1996; Diebold and Rudebusch, 1989): the spectrum of an $I(d)$ process has unbounded power as $\omega \rightarrow 0$ and concentrates extra power on any low-frequency band. We therefore rerun Gate 1 with the AR(1) null replaced by an ARFIMA(0, $\hat{d}$, 0) null. Simulation uses the Hosking (1981) fractional-difference convolution $(1 - L)^{-d} = \sum_k \psi_k L^k$ with $\psi_0 = 1$, $\psi_k = \psi_{k-1} \cdot (k - 1 + d)/k$ and Gaussian innovations whose variance matches the residuals of $(1 - L)^{\hat{d}} y_t$. Implementation: `ecowave/cycles/surrogate.py:arfima_bootstrap_null`; generator: `ecowave/cycles/surrogate_generators.py:arfima_surrogate_series`; estimator: `ecowave/cycles/surrogate_generators.py:estimate_d_gph`; script: `scripts/arfima_null_per_cell.py`; sidecar: `reports/arfima_null_per_cell.json`. The joint (AR(1), ARFIMA) verdict is the load-bearing Gate 1 when $|\hat{d}| > 0.1$ on the cell.

Appendix B.7. Band-edge sensitivity

The canonical bands are anchored to the prior empirical literature (Kitchin = [3, 5], Juglar = [7, 11], Kuznets = [15, 25], Kondratieff = [40, 60]). To quantify the sensitivity of Gate 1 verdicts to the band-edge

choice we rerun the gate on six perturbations per cycle: tighten/loosen on each edge by $\pm 1y$ (Kitchin/Juglar) or $\pm 2y$ (Kuznets/Kondratieff), plus widen/narrow variants. Script: `scripts/band_sensitivity.py`; sidecar: `reports/band_sensitivity.json`. The headline pass-rate is reported alongside the rates under each perturbation, so a reader can assess whether the substantive concentrations of Section 5 survive small relabelings of the canonical bands.

Appendix B.8. Rolling-window Gate 1

To map the temporal stability of each cycle on each variable, we also report Gate 1 verdicts on a grid of rolling windows of fixed length (default 50y for Kitchin/Juglar/Kuznets, 200y for the Kondratieff window-bound cells), with step 25y. Script: `scripts/rolling_window_gates.py`; sidecar: `reports/rolling_window_gates.json`. The output supports a heat-map of cycle presence across history; the substantive question is whether a cycle that passes on the full window is present in each sub-epoch or only on a subset (Belle Époque, Trente Glorieuses, post-2008).

Appendix B.9. Dual-null surrogate test

Null 1: AR(1) bootstrap..

- 1: Fit $\hat{\phi}, \hat{\mu}, \hat{\sigma}$ on y by OLS.
- 2: **for** $b = 1, \dots, B$ **do**
- 3: Draw $\epsilon_t^{(b)} \sim \mathcal{N}(0, \hat{\sigma}^2)$.
- 4: Simulate $y_t^{(b)} = \hat{\mu} + \hat{\phi}(y_{t-1}^{(b)} - \hat{\mu}) + \epsilon_t^{(b)}$.
- 5: Compute band-power P_b of $y^{(b)}$ on the canonical band.
- 6: **end for**

7: $p_1 = (1 + \#\{b : P_b \geq P_0\})/(1 + B)$.

Null 2: Phase-randomised surrogate (coherence test)..

- 1: Compute $\hat{y}_k = \mathcal{F}[y]_k$ (DFT).
- 2: **for** $b = 1, \dots, B$ **do**
- 3: Draw new phases $\theta_k^{(b)} \sim \text{Uniform}[0, 2\pi)$ with conjugate symmetry
 $\theta_{T-k}^{(b)} = -\theta_k^{(b)}$.
- 4: Compute $y^{(b)} = \mathcal{F}^{-1}[[\hat{y}_k] e^{i\theta_k^{(b)}}]$.
- 5: Compute a within-band phase-coherence statistic C_b (mean resultant length of the Hilbert phase angles inside the canonical band).
- 6: **end for**
- 7: $p_2 = (1 + \#\{b : C_b \geq C_0\})/(1 + B)$.

The phase-randomised surrogate preserves the empirical Fourier amplitudes exactly. Null 2 is therefore not a test of cycle existence in isolation; it tests whether the temporal coherence of the Hilbert phase within the band exceeds what an amplitude-preserving randomisation would produce. We thank the referee who insisted on this clarification and report the literature on the limitation of phase-randomised nulls for cycle detection ([Schreiber and Schmitz, 2000](#)).

Appendix B.10. Calibration on synthetic alternatives

We report empirical detection power of Gate 1 against three families of synthetic alternatives at five signal-to-noise ratios. For each combination we run $N = 1000$ replicates of panel length $T = 150$ (matching JST R6 annual length) and report the empirical rejection rate $\hat{\pi}$.

Family I: AR(1) + cosine.. $y_t = \phi y_{t-1} + \beta \cos(2\pi t/L) + \epsilon_t$ with L at the canonical band centre and β tuned so that the cosine variance contributes a fraction $\text{SNR} = \beta^2 / (\beta^2 + \sigma_\epsilon^2 / (1 - \phi^2))$ of total variance. This is the favourable case.

Family II: Two-regime MS-AR(1).. Hamilton-style model with mean-shift between regimes and dwell times sampled from a geometric distribution centred on the canonical band. SNR controls the mean separation.

Family III: Multifractal random walk (MFRW).. The MFRW of [Bacry et al. \(2001\)](#) with embedded cyclical modulation of the integrated intermittency. SNR controls the cyclical share of the variance.

Empirical power.. Table [B.3](#) reports the empirical detection power of the AR(1) bootstrap component of Gate [1](#) (the primary existence test). The numbers are from the actual run shipped in `reports/calibration_v1.json`.

Table B.3: Empirical detection power of the AR(1) bootstrap component of Gate [1](#) at $\alpha = 0.05$, on $T = 150$ annual observations, $N = 100$ replicates, $B = 200$ surrogates. Source: `reports/calibration_v1.json` (v1, period at band centre 8 y, band $[7, 11]$ y).

Generative model	SNR				
	0.2	0.3	0.5	1.0	2.0
Family I — AR(1) + cosine	0.82	1.00	1.00	1.00	1.00
Family II — MS-AR(1)	0.04	0.03	0.03	0.08	0.26
Family III — MFRW + modulation	0.10	0.07	0.06	0.02	0.01

Implication.. The empirical calibration is stark. On the simplest possible alternative—a cosine embedded in linear AR(1) noise—the AR(1) bootstrap is extremely powerful, with $\geq 80\%$ detection at $\text{SNR} \geq 0.2$ and saturation by $\text{SNR} \geq 0.3$. On the two realistic alternatives, however, the picture inverts. Against Markov-switching AR(1) processes whose mean-shift dwell time matches the canonical band, the test detects between 3% and 26% of cases across the SNR range. Against multifractal random walks with cyclical modulation of the intermittency, detection peaks at 10% at low SNR and *decreases* towards 1% at high SNR (large SNR amplifies the volatility-of-volatility, which the AR(1) bootstrap is not designed to recognise as a cycle).

This is the load-bearing caveat of the paper. The test we apply to the panels is well-suited to detect simple linear sinusoidal cycles but is poorly suited to detect the regime-switching and multifractal cyclical structures that the modern empirical macroeconomic literature has documented ([Hamilton, 1989](#); [Calvet and Fisher, 2002](#); [King et al., 1991](#)). The rejection of canonical periodicities in Section 5 therefore should be read as “no *linear sinusoidal* canonical periodicity passes the protocol”—which is the historical claim under attack ([Kitchin 1923](#); [Juglar 1862](#); [Kuznets 1930](#); [Kondratieff 1925](#) all presented their cycles as approximately sinusoidal regularities)—and *not* as a refutation of cyclical structure as such. We make this disclosure load-bearing rather than incidental.

The phase-coherence null in calibration.. The phase-coherence p -value reported by `phase_scramble_null` is, on these calibration panels, near-uniformly distributed on $[0, 1]$ (median across all calibration cells: 0.81 on

AR(1)+cosine, 0.49 on MS-AR(1), 0.50 on MFRW). The coherence statistic does not discriminate real cycles from spectrum-preserving surrogates on these generative models, consistent with the theoretical observation of [Schreiber and Schmitz \(2000\)](#) that band-power-based statistics against phase-randomised surrogates are degenerate by Parseval. The phase-coherence verdict is therefore reported as a diagnostic but is not the primary Gate 1.

Appendix B.11. False-positive calibration

On synthetic AR(1) panels ($\phi \in \{0.3, 0.6, 0.9\}$, $T \in \{60, 150, 300\}$) the empirical false-positive rate of Gate 1 at $\alpha = 0.05$ is, over $N = 1000$ replicates, 0.048 ± 0.007 . The size of the test is controlled. The synthetic study replicates the [Killick et al. \(2012\)](#) AR-coal-output benchmark exactly; the Bry-Boschan estimator on the original NBER business-cycle dating reproduces the published turning points; the Hamilton MS-AR(1) on US industrial production 1947–1995 reproduces [Hamilton \(1989\)](#)’s published transition matrix within EM tolerance. Calibration outputs are exported to `reports/calibration_v1.json`.

Appendix C. Acceptance criteria: full specification

This appendix records the protocol thresholds and their limitations.

Appendix C.1. “Threshold transparency”, not pre-registration

The cycle bands, surrogate counts, gate thresholds and safeguard parameters were committed to the project’s public Git history before any panel was ingested for analysis. The commitment is verifiable through dated Git tags and the immutable `MappingProxyType` export of `ecowave/cycles/bands`.

py:CYCLE_BANDS. We refer to this arrangement throughout the paper as *threshold transparency*.

We do *not* claim this is a formal pre-registration. Formal pre-registration as practised on the OSF or AEA RCT registries requires submission of a complete analysis plan to an independent third party that timestamps the document outside the researcher’s control. Our commitment is internal to the project repository: it is auditable but it is auditable by readers, not by an external registrar. We record the distinction explicitly because it matters for the interpretation of any negative result, and we avoid the word “pre-registered” in the body of the paper to prevent the conflation.

The threshold values themselves are anchored to the prior empirical literature on each cycle, not derived from the data analysed here. This is the only sense in which the protocol is “pre-data”.

Forward commitment: OSF for the out-of-sample extension.. We plan to register an out-of-sample replication protocol on the OSF before any post-2024 data update is ingested. The registered protocol will specify the cycle bands, surrogate counts, gate thresholds, safeguard parameters and panel definitions identically to the present paper, and will constrain the analyst from any post-hoc modification of those parameters on the new data. Future extensions of the present analysis will therefore be pre-registered in the formal sense.

Appendix C.2. Bands and anchor references

Cycle	Band (years)	Anchor references
Kitchin	3–5	Kitchin (1923) ; Wen (2005) ; Diebolt and Doliger (2008)
Juglar	7–11	Juglar (1862) ; Schumpeter (1939)
Kuznets	15–25	Kuznets (1930) ; Korotayev and Tsirel (2010)
Kondratieff	40–60	Kondratieff (1925) ; Modelski (1987) ; Korotayev and Tsirel (2010)

Appendix C.3. Gate 1 (existence-plus-coherence)

At significance $\alpha = 0.05$ on B AR(1) surrogates per cell ($B = 1000$ on JST R6, $B = 500$ on Bank of England Millennium and World Bank, $B = 300$ on the quarterly panel, $B = 200$ on the sectoral panel; the count is chosen to keep total runtime tractable on the project’s reproducible Docker image), the empirical band-power must exceed the 95th percentile of the AR(1)-bootstrap null *and* the within-band Hilbert phase-coherence statistic must exceed the 95th percentile of the Theiler phase-randomised null. The two surrogate p -values p_1 (existence) and p_{coh} (coherence) are recorded separately and reported in Section 5; this acknowledges that the two surrogates test different aspects of the data ([Schreiber and Schmitz, 2000](#)).

Appendix C.4. Gate 2 (method consensus): dependence of estimators

A cell passes Gate 2 iff at least three of four estimators (Christiano-Fitzgerald, Morlet, Markov-switching, PELT) concur on the dominant phase at every common time point. The four estimators are not statistically independent. On null AR(1) panels ($\phi = 0.5$, $T = 150$,

$N = 200$ replicates), the empirical pairwise phase agreement between Christiano-Fitzgerald-Hilbert and a Bry-Boschan-style sign-of-velocity proxy is 0.79 (mean; median 0.79; see `reports/estimator_pairwise_v1.json` and `scripts/measure_estimator_agreement.py`). This is an overestimate of inter-method independence for the full four-method consensus, but it provides the right order of magnitude: the estimators share enough of the same slope-following structure that the “3-of-4” rule cannot deliver the four-fold independence factor that would justify a Bonferroni-style multiplicity argument. We therefore report Benjamini-Hochberg adjusted p -values on the full grid of Gate 1 tests in Appendix [Appendix D](#) as a complementary robustness perspective, using the `ecowave.cycles.surrogate.benjamini_hochberg_adjust` utility shipped with the project.

Pre-registered relaxations.. Two consensus relaxations are declared in advance and grounded in sampling constraints: (i) two votes for Kitchen on annual data, because PELT and Bry-Boschan are under-powered when the cycle period is fewer than five times the sampling interval; (ii) two votes for Kondratieff on panels whose window covers fewer than two full periods (< 120 annual observations), because Markov-switching and PELT cannot deliver enough state changes. The relaxations are coded in `ecowave/cycles/consensus.py:RELAX_RULES`.

Appendix C.5. Gate 3 (cross-aggregate concordance): not a universality test

We count, on the WB income-stratified family (WLD, HIC, UMC, LMC, LIC), how many aggregates independently pass Gates 1–2. We *do not* require universal concordance: the modern theory of the business cycle predicts

that institutional, monetary and financial differences across income groups should produce differential cyclical responses (Stock and Watson, 2005). A cycle that passes on HIC and UMC but not on LMC or LIC is reported as “advanced-economy-specific”; this is informative, not disqualifying.

Verdicts: “cross-aggregate consistent” (5/5 or 4/5), “advanced-economy specific” ($\text{HIC} \cap \text{UMC}$, others fail), “regional” ($\leq 2/5$), “rejected” (0/5).

Appendix C.6. Per-variable safeguard: threshold sensitivity

The threshold of 0.5—the fraction of component variables required to individually pass Gate 1 for a composite to be admitted—is taken from Wen (2005)’s usage on inventory data. We report sensitivity in Appendix Appendix D: the qualitative verdict (no canonical cycle passes the cascade on more than a minority of panels) is invariant for thresholds in $[0.3, 0.7]$. On panels with few component variables (BoE Millennium individual UK series; sectoral 11-series panel) the threshold becomes mechanical and we discuss the implications in Appendix Appendix D.

Appendix C.7. Multiple-comparison correction: BH-FDR as robustness

We do not apply Bonferroni or Holm correction across the $\sim 1\,200$ cells of the grid as a primary verdict because the cascade itself provides multiplicity protection. However, because the cascade protection rests on partial independence assumptions that the empirical correlations of Section 3 only partially satisfy, we report Benjamini-Hochberg (Benjamini and Hochberg, 1995) adjusted p -values on the full Gate 1 grid as a complementary perspective. The verdict is qualitatively invariant; the few cells whose status changes are listed in Appendix Appendix D.

Appendix C.8. Conditions that would falsify the protocol itself

The protocol is a conjecture. We declare it falsified on observation of any of the following:

- (i) Empirical false-positive rate at Gate 1 on synthetic AR(1) panels exceeds 0.05 by more than two standard errors over $N = 1000$ replicates.
- (ii) Power against Family I synthetic alternatives at $\text{SNR} = 1.0$ falls below 0.80 (Section [Appendix B.10](#)).
- (iii) Any of the four Gate 2 estimators fails to reproduce its canonical published result on the corresponding textbook benchmark.

All three conditions are routinely checked by the project’s test-suite (`tests/`). Calibration outputs are in `reports/calibration_v1.json`.

Appendix D. Per-variable evidence, sensitivity, and FDR-adjusted verdict

This appendix reports the per-variable Gate 1 evidence that underpins Section 5, with raw p -values rather than pass/fail indicators, with sensitivity to the safeguard threshold and to the analysis window, and with the Benjamini-Hochberg false-discovery-rate adjusted verdict as a robustness check.

Appendix D.1. Sectoral panel: empirical p -values

Table [D.4](#) reports the AR(1) bootstrap p -value p_1 of Gate 1 for every (variable, cycle) cell of the sectoral panel, computed by `scripts/wen_replication.py` on the raw data shipped with the project. The Kondratieff column is empty because no sectoral series in the panel is long enough

Table D.4: Empirical [Wen \(2005\)](#) replication on the sectoral panel. AR(1) bootstrap p_1 on $B = 200$ surrogates per cell, computed from the raw CSV by `scripts/wen_replication.py`; source: `reports/wen_replication_per_variable.json`. Cells with $p_1 < 0.05$ are shown in bold.

Variable (panel period)	Kitchin	Juglar	Kuznets	Kondratieff
UK coal output (1700–2019)	0.811	0.995	0.632	—
US coal output (1856–1958)	0.224	0.811	0.726	—
US cotton (1912–1961)	0.169	0.975	—	—
US industrial prod. (1919–2026)	0.174	0.224	0.881	—
US rail freight (1866–1922)	0.632	0.761	—	—
US steel (1863–1965)	0.766	0.189	—	—
US wheat (1866–1952)	0.040	0.592	—	—
US WPI (1860–2026)	0.299	0.015	0.701	—
World coal (1900–2024)	0.299	0.672	0.020	—
World oil (1900–2024)	0.701	0.488	0.274	—
Survival rate at $\alpha = 0.05$	1/10	1/10	1/6	0/0
Median p_1	0.299	0.632	0.667	—

to support a meaningful test on a 40–60y band (we require $N > 4 \cdot 60 = 240$ annual observations); a dash indicates an untestable band.

The unadjusted survival rate is 3 of 26 testable cells at $\alpha = 0.05$ (US wheat Kitchin, US WPI Juglar, world coal Kuznets). Under Benjamini-Hochberg adjustment at FDR 0.05 applied to the 26 p_1 values, the critical threshold is $p^* = 0.05 \cdot 1/26 \approx 0.0019$; none of the three individually-significant cells passes the BH threshold, so zero of twenty-six cells survives the FDR-adjusted gate. The three positive cells under the unadjusted gate are individually

plausible exceptions (US wheat Kitchin: agricultural inventory cycle; US WPI Juglar: the original Juglar variable; world coal Kuznets: long-swing residential-construction-style pattern) but do not constitute evidence for a universal canonical periodicity. The result reproduces, on real historical data, the substantive finding of [Wen \(2005\)](#): cycles that appear on composite aggregates are typically not detectable on the underlying sectoral series.

Appendix D.2. Sensitivity to the analysis window

We re-run Gate 1 on each panel under two robustness windows. The qualitative verdict on the sectoral panel (3 unadjusted cells, 0 surviving BH-FDR) is invariant. On JST R6 the Kondratieff band is window-bound (maximum 151 annual observations per series, below the threshold of 240 required for a meaningful 40–60y test) and therefore not part of the window-sensitivity analysis; the JST verdict on the Juglar and Kuznets bands is invariant to restricting the panel to 1870–2007. On BoE Millennium, the three unadjusted positives (UK debt/GDP on Kuznets and Kondratieff, UK unemployment on Juglar) are stable under restriction to 1700–1945. On the quarterly panel, the Kitchin and Juglar positives are stable under restriction to the pre-2007 sub-window. The substantive concentrations of Section 5 therefore do not depend on the inclusion of the post-2008 period.

Appendix D.3. Per-variable Kondratieff on BoE Millennium

Table D.5 reports the AR(1) bootstrap p_1 of Gate 1 for the ten Bank of England Millennium variables long enough to support a Kondratieff (40–60y) test ($N > 240$ annual observations). The numbers come from the raw long-format CSV (`data_raw/boe/annual.csv`) via `scripts/`

boe_kondratieff_per_variable.py; source: reports/boe_per_variable.json.

Table D.5: Per-variable Kondratieff test on the Bank of England Millennium panel, 1700–2016. p_1 on $B = 500$ AR(1) surrogates. Cells with $p_1 < 0.05$ shown in bold.

Variable (panel period)	p_1	Gate 1
Real GDP (1700–2016)	0.643	rejected
Composite real GDP, long (1700–2016)	0.669	rejected
CPI (1700–2016)	0.615	rejected
Wholesale price index (1700–2016)	0.212	rejected
Bank Rate (1700–2016)	0.778	rejected
Consols long-term yield (1703–2016)	0.102	rejected
Public debt / GDP (1700–2016)	0.004	passes
Share prices (1700–2016)	0.986	rejected
Population GB/NI (1700–2016)	0.291	rejected
Unemployment rate (1760–2016)	0.088	rejected
Survival rate at $\alpha = 0.05$		1/10

One of ten Kondratieff-eligible BoE variables passes Gate 1 at the unadjusted significance level: the public-debt-to-GDP ratio, with $p_1 = 0.004$. This is the canonical Reinhart-Rogoff long-debt-cycle signature, driven on UK data by the Napoleonic, Crimean and two World Wars financing peaks. The same variable also passes the Kuznets band ($p_1 = 0.006$). Both passings are consistent with prior literature on UK fiscal cycles (Reinhart and Rogoff, 2009) and are properly understood as *debt-financing* cycles tied to exogenous war shocks, not as evidence for the existence of a universal endogenous 40–60y

periodicity.

Across the full BoE Gate 1 run (16 variables $\times$ 4 cycle bands, 58 testable cells), three cells pass unadjusted: public debt / GDP on Kuznets ($p_1 = 0.006$) and Kondratieff ($p_1 = 0.004$), and the unemployment rate on Juglar ($p_1 = 0.006$). Under Benjamini-Hochberg adjustment at FDR 0.05 (critical threshold $p^* = 0.05/58 \approx 0.00086$), none of the three survives. The expected number of unadjusted positives under the joint null is 2.9; observing three is consistent with sampling noise. The unadjusted positives are nonetheless interesting individually and are flagged for the substantive literature, but they do not constitute evidence for the strong universalist reading of any canonical cycle.

Appendix D.4. Per-variable Juglar and Kuznets on Jordà-Schularick-Taylor R6

We apply Gate 1 to every (country, variable, cycle) cell of the JST R6 panel for which the panel window admits a meaningful test ($N > 4 \cdot \text{hi_years}$). The Kondratieff band cannot be evaluated on JST: the maximum window 1870–2020 yields at most 151 annual observations per series, short of our threshold $4 \cdot 60 = 240$. The Kondratieff verdict on the longest panels therefore rests on BoE Millennium alone. The script that produces the numbers below is `scripts/jst_per_variable.py`; source: `reports/jst_per_variable_b1000.json` ($B = 1000$ surrogates).

Juglar (605 testable cells).. 67 cells pass Gate 1 at the unadjusted significance level (11.1%), against an expectation of approximately 30 under the joint null at $\alpha = 0.05$ —a 2.2-fold excess. The distribution of the positives across

variables is substantively informative: LH_XRUSD (bilateral USD exchange rate) passes on 11 of 18 countries (61 %); LH_INV (investment/GDP) on 7 of 18 (39 %); LH_UNRATE on 6 of 18 (33 %); LH_BUSCREDIT on 5 of 15 (33 %); and LH_RCONS, LH_HPI, LH_RGDP_BARRO, LH_DEBTGDP on 4 each. The smallest p_1 ($= 0.001$, the surrogate floor at $B = 1000$) is observed on investment in Switzerland and Canada and on unemployment in Portugal.

Kuznets (529 testable cells).. 51 cells pass Gate 1 at the unadjusted significance level (9.6%), against an expectation of ~ 26 —a 1.9-fold excess. LH_HPI (house prices) passes on **6 of 13 countries (46 %)**; LH_POP (population) on 7 of 18 (39 %); LH_CREDIT (total loans) on 7 of 17 (41 %); LH_MORT on 5; LH_DEBTGDP and LH_CA on 4. The smallest p_1 ($= 0.001$) is observed on population in Belgium and Portugal and on house prices in the Netherlands and the USA.

Benjamini-Hochberg adjustment on the JST grid.. Across the full 1134-cell JST grid (Juglar + Kuznets), the BH-FDR critical threshold at $\alpha = 0.05$ is $p^* = 0.05/1134 \approx 4.4 \cdot 10^{-5}$. With $B = 1000$ surrogates, no individual cell can attain a p_1 below the surrogate floor $1/(B + 1) \approx 1 \cdot 10^{-3}$, which is itself one order of magnitude above p^* . The 118 unadjusted positives therefore all fail BH adjustment in the present run; whether some of them would survive at $B \geq 10\,000$ is an open empirical question, and the `scripts/jst_per_variable.py` script supports such a run.

Interpretation.. The unadjusted JST positives are substantively interesting and not noise: they concentrate exactly where the canonical literature predicts they should (investment for Juglar; housing, demographics and credit for

Kuznets). They do not, however, amount to a universal canonical cycle in the strong sense: the fraction of cells passing is small (10 %), the concentration is variable-specific rather than universal across the macro grid, and Benjamini-Hochberg correction at FDR 0.05 erases the collective claim. The JST evidence is consistent with the substantive theory of Juglar (investment-financed cycles) and Kuznets (construction swings) but inconsistent with a strong universalist sinusoidal reading on every variable.

Appendix D.5. Sensitivity to the safeguard threshold

For thresholds in $\{0.3, 0.5, 0.7\}$, the qualitative verdict on every panel is invariant. The only cell whose status depends on the threshold is the WB world Kitchen composite (rejected at any threshold ≥ 0.3 under the safeguard; admitted only if no safeguard is applied). We retain 0.5 as the primary value following [Wen \(2005\)](#)’s usage.

Appendix D.6. Long-memory and stationarity diagnostics

Following the R1 referee recommendation, we report on every cell the four pre-cycle diagnostics introduced in Appendix [Appendix B.5](#): ADF ([Dickey and Fuller, 1979](#)), KPSS ([Kwiatkowski et al., 1992](#)), GPH fractional-integration order $\hat{d}$ ([Geweke and Porter-Hudak, 1983](#)) and DFA Hurst exponent $\hat{H}$ ([Peng et al., 1994](#)). The script is `scripts/long_memory_diagnostics.py`; sidecar: `reports/long_memory_diagnostics.json`. Diagnostics are computed on the input series as ingested (without the `log_diff/share_of_gdp` transformation that Gate 1 applies subsequently), to characterise the raw long-memory profile of each variable.

Table D.6: Long-memory and stationarity diagnostics on the raw input series. ADF non-rejection signals a unit root; KPSS rejection signals departure from stationarity; $|\hat{d}| > 0.1$ or $\hat{H} > 0.6$ signal long memory beyond AR(1). Numbers come from `reports/long_memory_diagnostics.json`.

Diagnostic	JST R6 (611 cells)	BoE (65 cells)
Median $\hat{d}_{\text{GPH}}$	+0.490	+0.490
$ \hat{d} > 0.1$ (share)	97 %	100 %
$ \hat{d} > 0.4$ (share)	88 %	95 %
Median $\hat{H}_{\text{DFA}}$	1.76	1.64
$\hat{H} > 0.6$ (share)	97 %	94 %
$\hat{H} > 1.0$ (share)	88 %	92 %
ADF unit-root not rejected at $\alpha = 0.05$	81 %	83 %
KPSS stationarity rejected at $\alpha = 0.05$	77 %	89 %

Implication.. On both panels the raw input series exhibit substantial long-memory and near-unit-root behaviour: median $\hat{H} \geq 1.6$ (consistent with I(1) random-walk-like dynamics on level series), $|\hat{d}|$ saturated at the GPH clipping bound of 0.49 on the majority of cells, and ADF unable to reject the unit-root null on roughly four cells in five. An AR(1) null fitted on the raw level is therefore mis-specified for the majority of cells. Gate 1 in practice operates on the post-transformation series (`log_diff` for most level series, `share_of_gdp` for ratios); after log-differencing the persistence reduces materially. The honest reading is nonetheless that the AR(1) Gate 1 verdict should be cross-validated against the ARFIMA(0, $\hat{d}$,0) Gate 1 verdict reported by `scripts/arfima_null_per_cell.py` (sidecar `reports/`

`arfima_null_per_cell.json`). Cells that pass under both nulls gain credibility; cells that pass only the AR(1) null and fail the ARFIMA null should be downgraded as candidates for the variable-specific vindication claim.

Appendix D.7. Band-edge sensitivity

Following the R4 recommendation, we rerun Gate 1 with the canonical bands perturbed on each edge by $\pm 1y$ (Kitchin/Juglar) or $\pm 2y$ (Kuznets/Kondratieff), plus widen/narrow variants. Script: `scripts/band_sensitivity.py`; sidecar: `reports/band_sensitivity.json`. A substantive cycle should remain detected when the band is perturbed by one year; a band-fitting artefact will not. Table D.7 reports the BoE Millennium pass-rates (1541 cells, $B = 300$ surrogates, 65 variables, 27 cycle $\times$ perturbation aggregates).

Interpretation.. Three of the four cycles are stable to band-edge perturbation: Kuznets (9.4–14.5 %), Kondratieff (7.9–14.7 %) and Juglar (9.2–15.4 %) span a range of at most six percentage points around the anchor pass-rate. **Kitchin is the exception:** tightening the lower edge to [4,5] yields a zero pass-rate, while loosening the upper edge to [3,6] more than doubles the pass-rate to 16.9 %. The asymmetric sensitivity is exactly the signature of a band-fitting artefact: the substantive signal on BoE within a [3,5] window does not survive a one-year perturbation in either direction. This vindicates the R4 critique on the Kitchin cell of BoE specifically, and we discount the BoE Kitchin headline accordingly in Section 5.4. The JST and quarterly panels, which carry the load-bearing Kitchin claims of the paper, should be re-run with the same sensitivity script before any Kitchin-specific revision; the BoE Kitchin signal

Table D.7: Band-edge sensitivity of Gate 1 on BoE Millennium. Pass-rate by (cycle, perturbation). “anchor” is the literature-anchored band. Perturbations move one edge in or out by $\pm 1y$ (Kitchin/Juglar) or $\pm 2y$ (Kuznets/Kondratieff); “narrow” and “widen” move both edges simultaneously. Source: `reports/band_sensitivity.json`.

Cycle	anchor	tighten lo	loosen lo	tighten hi	loosen hi	narrow	widen
Juglar	15.4 %	10.8 %	12.3 %	15.4 %	9.2 %	13.8 %	10.8 %
[7,11]							
Kuznets	12.9 %	11.3 %	14.5 %	10.9 %	11.3 %	9.4 %	12.9 %
[15,25]							
Kondratieff	10.5 %	7.9 %	10.5 %	13.2 %	11.8 %	13.2 %	14.7 %
[40,60]							
Kitchin	7.7 %	9.2 %	0.0 %	4.6 %	16.9 %	4.6 %	4.6 %
[3,5]							

is no longer treated as supporting evidence.

Appendix D.8. Rolling-window Gate 1: temporal stability

Following the R5 recommendation, we report Gate 1 verdicts on a grid of rolling windows of fixed length (80y for the Kondratieff window-bound cells; 50y for Kitchin/Juglar/Kuznets; step 40y on annual panels). Script: `scripts/rolling_window_gates.py`; sidecar: `reports/rolling_window_gates.json`. On the BoE Millennium 1700–2016 the run produces 1 646 rolling cells across 65 variables; aggregate pass-rates per cycle are:

- Juglar: 57 / 466 windows pass Gate 1 (12.2 %)
- Kitchin: 40 / 466 (8.6 %)

- Kuznets: 39 / 430 (9.1 %)
- Kondratieff: 40 / 284 (14.1 %)

These rates are modestly elevated above the 5 % nominal null but compatible with intermittent presence rather than stationary cyclicalities across the full 1700–2016 horizon. A substantive cycle that lives in one epoch but is absent in another (e.g. Belle Époque vs Trente Glorieuses vs post-2008) is not detected by a single full-window test; the heat-map shipped in the JSON sidecar resolves the epoch-by-epoch presence.

Appendix D.9. ARFIMA-vs-AR(1) Gate 1 on BoE Millennium

Following the R1 recommendation, we rerun Gate 1 on every BoE cell with the AR(1) null replaced by an ARFIMA(0, $\hat{d}_{\text{GPH}}$, 0) null. The cross-tabulation of the two verdicts is the empirical test of whether the AR(1) positives survive long-memory-aware re-evaluation. Script: `scripts/arfima_null_per_cell.py`; sidecar: `reports/arfima_null_per_cell.json`; on BoE (230 cells total across the four cycles, $B = 500$).

Table D.8: Cross-tabulation of Gate 1 verdicts under AR(1) versus ARFIMA(0, $\hat{d}$, 0) nulls on the BoE Millennium panel.

Cycle	n	AR(1) pass	ARFIMA pass	both	only AR(1)
Kitchin	65	5	1	1	4
Juglar	65	8	4	2	6
Kuznets	62	8	9	4	4
Kondratieff	38	5	6	3	2

Interpretation.. Ten cells pass *both* the AR(1) and the ARFIMA null on BoE; they are the long-memory-robust subset of the BoE positives. Among these survivors the substantive headlines remain intact: **UK unemployment on Juglar** ($p_{1,AR(1)} = 0.004$, $p_{1,ARFIMA} = 0.002$), **UK central-government gross debt on Kondratieff** ($p_{1,AR(1)} = 0.032$, $p_{1,ARFIMA} = 0.048$), **UK public-sector debt on Kondratieff** ($p_{1,AR(1)} = 0.002$, $p_{1,ARFIMA} = 0.022$, $\hat{d} = +0.436$), **real USD-GBP exchange rate on Juglar**, **real-eri and nominal-eri on Kuznets**. The **Kondratieff debt-chronology reading of Section 5.5** is reinforced, not overturned, by the ARFIMA-conditional verdict: the headline p_1 value remains below 0.05 under both nulls, with a GPH-estimated $\hat{d}$ at the long-memory clipping bound (0.44–0.47) that confirms the series is fractionally integrated. The 16 cells that pass only the AR(1) null and fail the ARFIMA null are downgraded as candidates for the variable-specific vindication claim.